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VALIDATION TEST REPORT

FOR

FLOW GRIND TEST STANDS COMPUTER DATA ACQUISITION AND MONITOR SYSTEM used on SERVOVALVES

Prepared for

Woodward HRT

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Prepared by: Alex Moshirpour

Date: 23 March 2015

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REVISION PAGE

REVISION LETTER	DATE OF REVISION	REVISED PAGES	REVISIONS
A	05/14/15	Title Pg	Revised to reflect current revision.
		2	Para. 1.1 – WAS: "... This report compares ...SWCD Version -04 ..." IS: "This revision ... SWCD Version -06 ..." ADDED new para. 1.2 and renumbered subsequent paragraphs.
		5	Para. 3.2 – WAS: "... Version -04." IS: "... Version -06."
		8	Para. 4.5 – WAS: "... Appendix A." IS: "... Appendix A for SWCD 97900503, Version -04."
		37	Appendix A title page – WAS: "... Data Plots" IS: "... Data Plots for SWCD 97900503 Version -04"

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REVISION PAGE

REVISION LETTER	DATE OF REVISION	REVISED PAGES	REVISIONS
B	06/19/15	Title Pg	Revised to reflect current revision.
		2	Para. 1.1 – WAS: "... SWCD Version -04 software with Version -06 ..." IS: "... SWCD Version -06 software with Version -07 ..." DELETED sentence: "The Version -04 ... counting." WAS: "... SWCD Version -06 ..." IS: "... SWCD Version -07 ..." REPLACED para. 1.2
		5	Para. 3.2 – WAS: "... Version -06." IS: "... Version -07."
		8	Para. 4.5 – WAS: "... Version -04." IS: "... Version -04, which are ... -07 software."

Prepared By: Alex Moshirpour Date: 19 June 2015

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1.0 INTRODUCTION

1.1 Objective

The objective of this Woodward HRT (WHRT) document is to describe the results of the testing performed for validation of the test stand that is used to verify the specification requirements for flow grind spool and sleeve assemblies for various WHRT servovalves. The Flow Grind Test Stands (FGTS) use Software Control Drawing (SWCD) 97900503 to program the Flow Grind Digital XY plotter. This revision of the validation is issued to replace SWCD Version -06 software with Version -07 software (see para. 1.2). After approval of this validation report, the SWCD Version -07 will be used on the FGTS and released to production.

1.2 Software Control Drawing Version Change

After releasing SWCD 97900503, Version -06, a change to the software was implemented that did not affect how the analog and encoder data are acquired or analyzed by the software. This change resulted in the updated SWCD 97900503 to Version -07. The changes were limited to a function that is making sure analysis results are presented as “blanks” for those parts that did not have overlays.

Since modifications to software Versions -06 and -07 were considered as cosmetic changes, the validation data collected using SWCD 97900503, Version -04, is still valid for Versions -06 and -07 software. Therefore, the analysis presented in this document uses the data obtained using SWCD 97900503, Version -04.

1.3 Conclusion

The results of the analysis compared the flow grind test data for four different servovalves tested on FGTS, and are presented in Figures 1 through 24 and Tables 1 through 4. The analysis results show that all tested parameters meet the acceptability criteria as described in Section 4.0; therefore, it is recommended that the FGTS with automated counting method be released to WHRT production.

1.4 Acronyms/Abbreviations

ATP	Acceptance Test Procedure
ATS	Automated Test Stand
DAQ	Data Acquisition
FGTS	Flow Grind Test Stand
StDev	Standard Deviation
SWCD	Software Control Drawing
WHRT	Woodward HRT

2.0 APPLICABLE DOCUMENTS

The following documents form a part of this report to the extent specified herein. Unless otherwise specified, the latest issue of the applicable documents specified shall be used.

2.1 Industry Documents

ANSI/NCSL-Z540.3	Requirements for the Calibration of Measuring and Test Equipment
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2.2 Woodward HRT Documents

2-PLC-00068	CM Procedure for Software Work Products
3-QCI-00181	Calibration System Procedure
28005087	Drawing, Flow Grind Test Stands
88004316	Test Software Requirements Matrix for Flow Grind Test Stands
97900503	Software Control Drawing for Flow Grind Digital XY Plotter Application
HR70700254	Calibration Procedure for Computer Data Acquisition for Flow Grind Test Stands

HRQP 2.200

Software Control Procedure

HRQP 7.001

Contamination Control Procedure

P05-20
Rev O-2

Product Acceptance Software

3.0 GENERAL REQUIREMENTS

3.1 Definition of Task

The objective of this report is to perform analysis to compare the flow grind test data for four different servovalve spool and sleeve assemblies tested on FGTS. The four spools and sleeves are used on four different servovalves identified as:

- a. P/N 22282790-104
- b. P/N 22284090-104
- c. P/N 22284250-104
- d. P/N 22253726 (25A)

The above servovalves were selected because they represent a cross section of the WHRT servovalves in production.

After the results of the analysis are reviewed and deemed acceptable, the FGTS will be released to production.

3.2 Item Description

The FGTSs are based on WHRT drawing 28005087 and parts list PL28005087. The test stands use National Instruments (NI) PCI-6289 Multifunction Data Acquisition (DAQ) card and NI-DAQmx driver. The software runs on the Microsoft MS Windows 7 operating system. Software revisions are identified by the WHRT SWCD 97900503, and access to

the software is controlled by a password. The FGTSs use updated SWCD 97900503,
Version -07.

3.3 Test Conditions

Test conditions were as specified in the applicable documents for the respective
servovalve.

3.4 Test Equipment Required

Test equipment was as specified in the applicable documents for the respective servovalve
and the Calibration Procedure HR70700254.

3.5 Test Setup

Test setup was as specified in the applicable documents for the respective servovalve.

3.6 Calibration Data

The FGTS was calibrated prior to validation testing in accordance with the Calibration
Procedure HR70700254 that is in compliance with WHRT Calibration System Procedure,
3-QCI-00181, which is in accordance with ANSI/NCSL-Z540.3.

3.7 Test Sequence

The test sequence was as specified in the applicable documents for the respective
servovalve, and included those tests that were related to the flow grind operation.

4.0 TEST RESULTS

Each servovalve listed in para. 3.1 was tested five times (sample size of five) on the FGTS
for the test parameters that were related to the flow grind operation. See Tables 1 through
4 for test result analysis for each servovalve for “Automated Counting” versus “Manual
Counting” methods.

WHRT Manufacturing Engineering had established three criteria for the acceptability of the analysis results as follows:

1. When the standard deviation (StDev) for “Automated Data” is less than or equal to the StDev of the “Manual Data”, and it is small relative to the tolerance band, the results for “Automated Data” will be acceptable.
2. If criterion 1 is not met, but the percent mean difference between the “Automated Data” and the “Manual Data”, with respect to the tolerance, is less than or equal to 20%, the results for “Automated Data” will be acceptable.
3. If criteria 1 and 2 are not met, then the criterion of “proceed at risk” will be allowed since the Acceptance Test Procedure (ATP) requirements at the end item (servo-valve) level will detect any nonconformance related to the flow grind operation.

In the following paragraphs, Figures 1 through 24 present the Boxplots and Descriptive Statistics results for comparison of the “Automated Data” and “Manual Data” distributions, and Tables 1 through 4 show test data analysis results.

4.1 Test Results for P/N 22282790-104

As shown in Figures 1 through 6 and Table 1, out of six parameters tested, five had low % mean difference with respect to the tolerance band as compared to 20% requirement. For the No-load Drift, the % mean difference was not applicable since no specification limits were required.

For two parameters, the differences between StDevs were negligible, and for one parameter the “Automated” StDev was lower. The StDev difference for the Lap Condition at 0.002/in. was zero, and for the neutral pressure there was no variation for both data sets. The “Automated” StDev for the pressure gain was higher but the criterion 2 was satisfactory.

Based on criteria 1 and/or 2, with the additional safeguard from criterion 3, the upgraded FGTS met the flow grind specification requirements with the “Automated Counting” method.

4.2 Test Results for P/N 22284090-104

As shown in Figures 7 through 12 and Table 2, out of six parameters tested, five had low % mean difference with respect to the tolerance band as compared to 20% requirement. For the No-load Drift, the % mean difference was not applicable since no specification limits were required.

For three parameters, the differences between StDevs were negligible, and for one parameter the “Automated” StDev was lower. For the No Load Drift the “Automated” StDev was higher.

Based on criteria 1 and/or 2, with the additional safeguard from criterion 3, the upgraded FGTS met the flow grind specification requirements with the “Automated Counting” method.

4.3 Test Results for P/N 22284250-104

As shown in Figures 13 through 18 and Table 3, out of six parameters tested, all had low % mean difference with respect to the tolerance band as compared to 20% requirement.

For four parameters, the differences between StDevs were negligible, and for the neutral pressure there was no variation for both data sets. The “Automated” StDev for the pressure gain was higher but the criterion 2 was satisfactory.

Based on criteria 1 and/or 2, with the additional safeguard from criterion 3, the upgraded FGTS met the flow grind specification requirements with the “Automated Counting” method.

4.4 Test Results for P/N 22253726 (25A)

As shown in Figures 19 through 24 and Table 4, out of five parameters tested, four had low % mean difference with respect to the tolerance band as compared to 20% requirement. For the No-load Drift, the % mean difference was not applicable since no specification limits were required. For the pressure gain the mean difference was much higher than the 20% requirement.

For three parameters, the differences between StDevs were negligible, and for the neutral pressure there was no variation for both data sets. For No-load Drift, the “Automated” StDev was higher. For pressure gain, the “Automated” StDev was much lower, and it met the criterion 1 requirement.

Based on criteria 1 and/ or 2, with the additional safeguard from criterion 3, the upgraded FGTS met the flow grind specification requirements with the “Automated Counting” method.

4.5 Test Data

The plots for all test data are included in Appendix A for SWCD 97900503, Version -04, which are also applicable to Versions -06 and -07 software.

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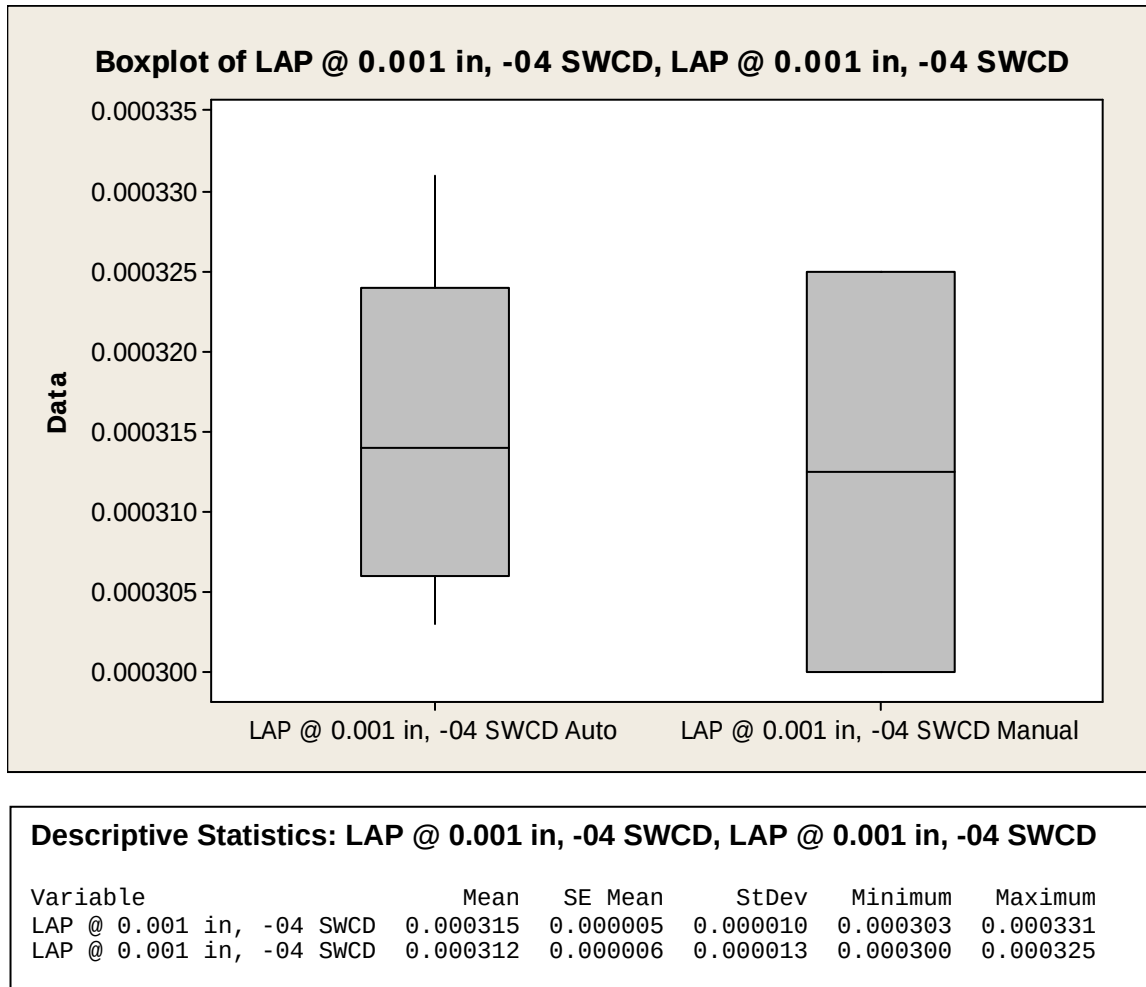


Figure 1. Boxplot of Lap at 0.001 in. for Servovalve P/N 22282790-104

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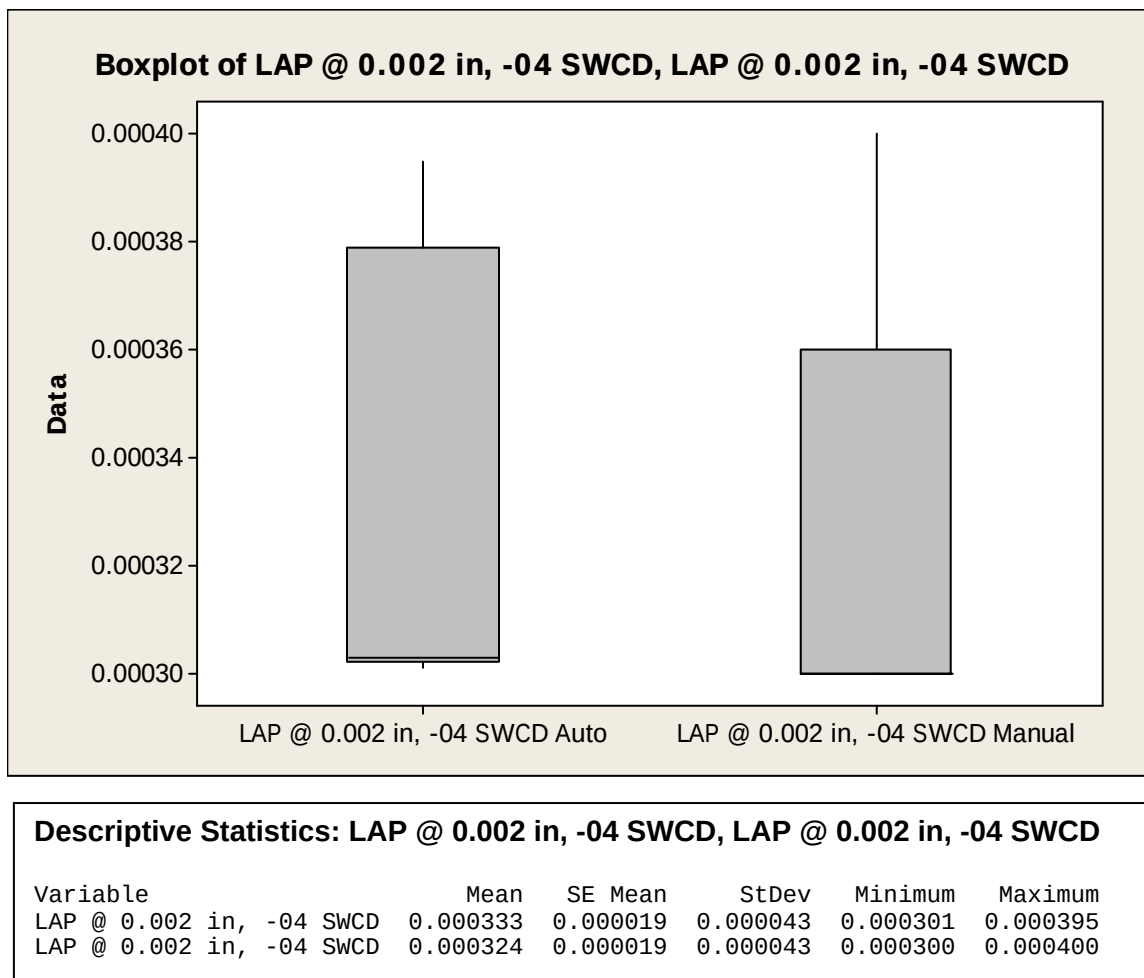


Figure 2. Boxplot of Lap at 0.002 in. for Servovalve P/N 22282790-104

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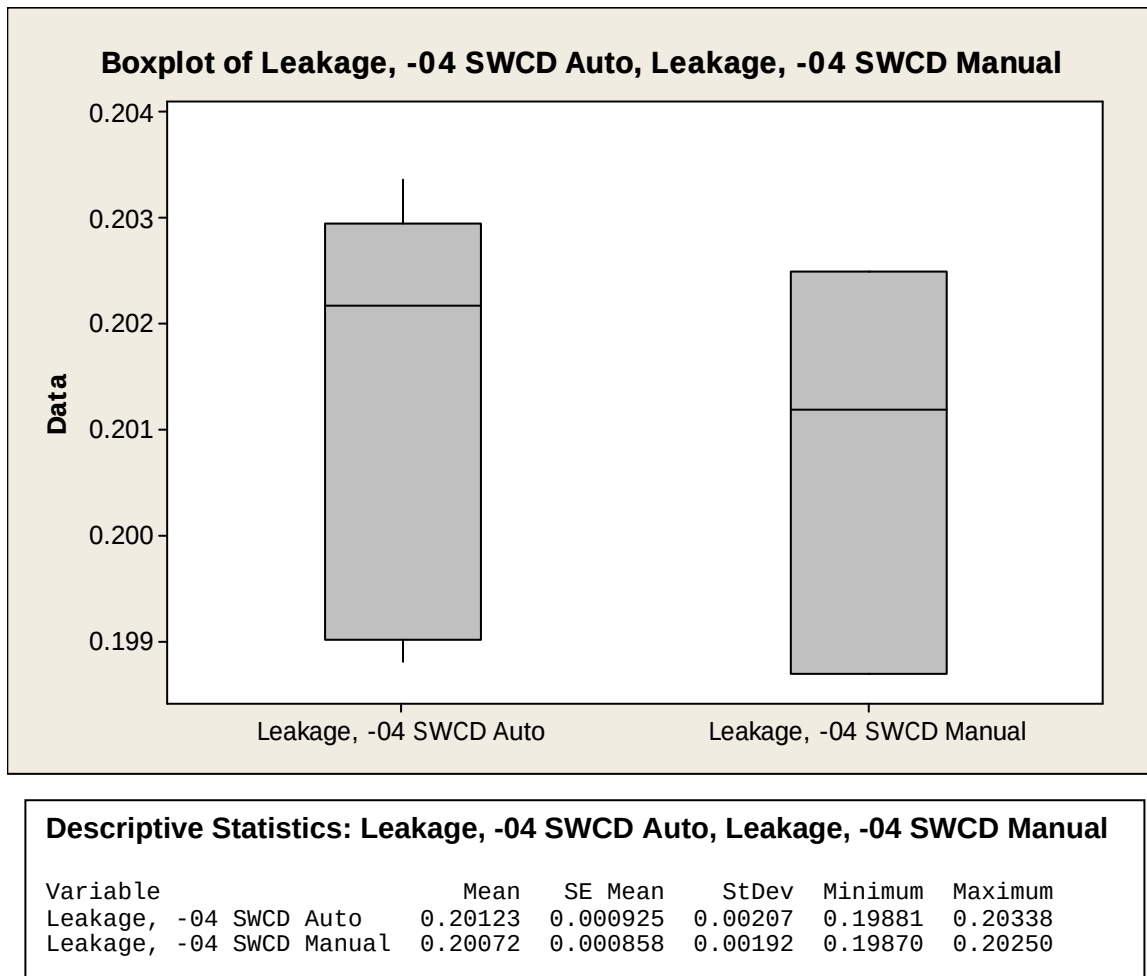


Figure 3. Boxplot of Leakage for Servovalve P/N 22282790-104

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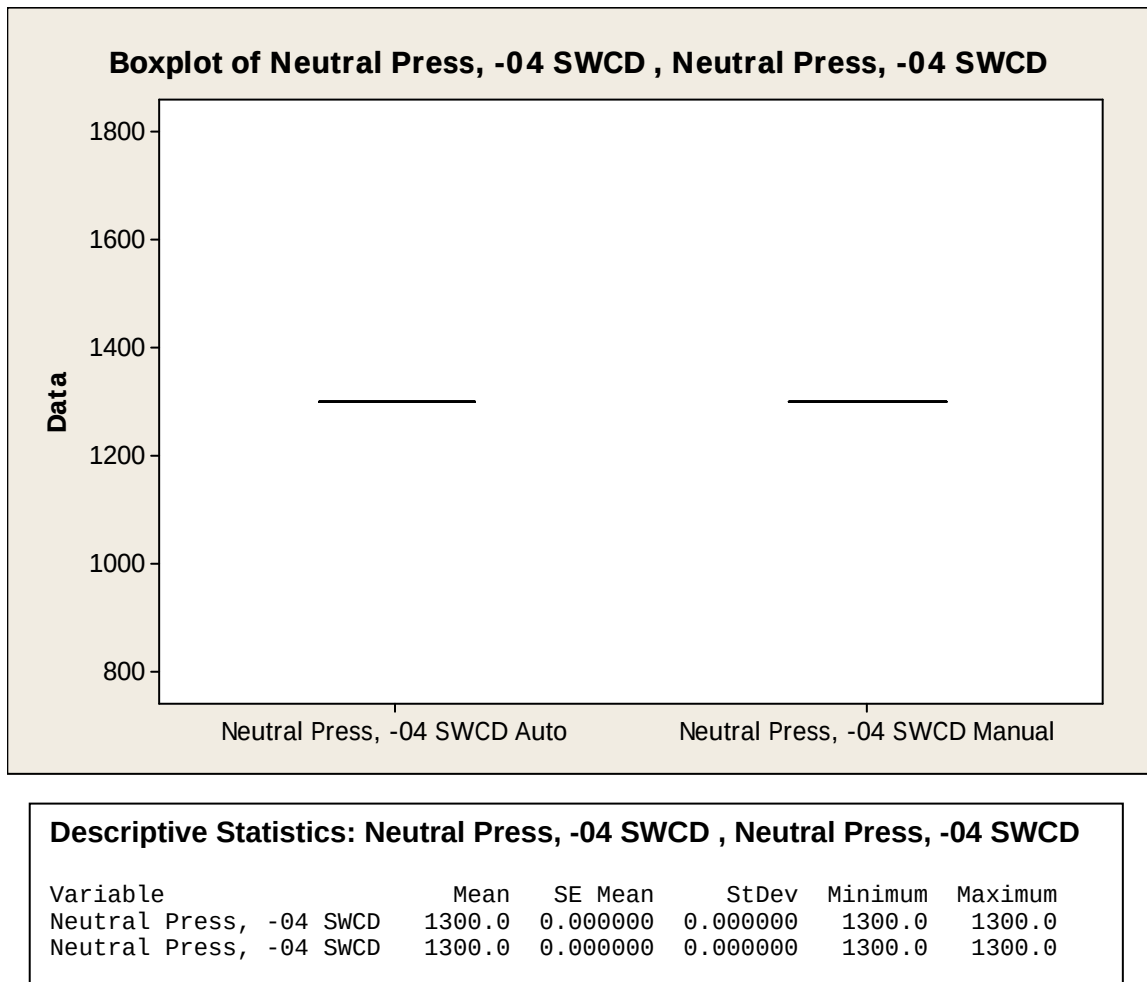


Figure 4. Boxplot of Neutral Press for Servovalve P/N 22282790-104

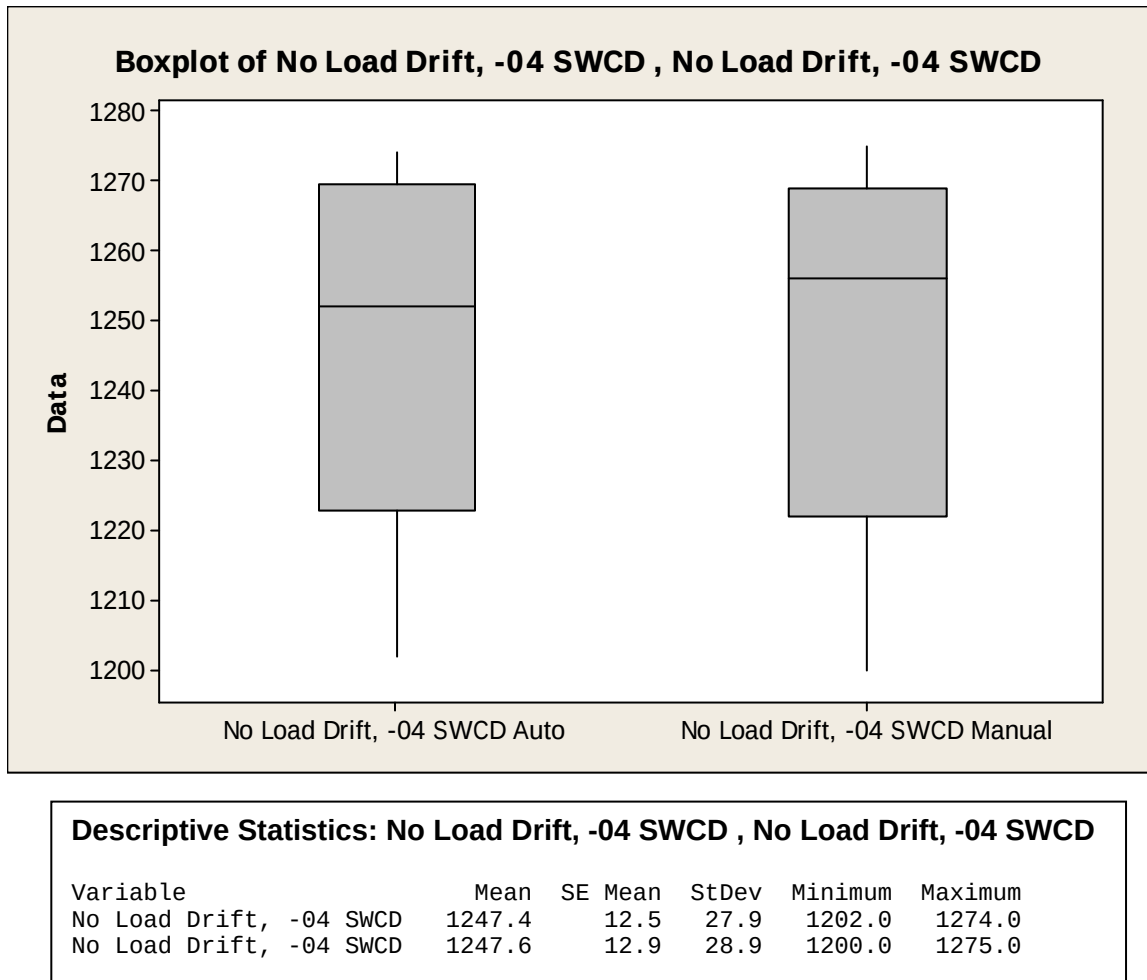


Figure 5. Boxplot of No-load Drift for Servovalve P/N 22282790-104

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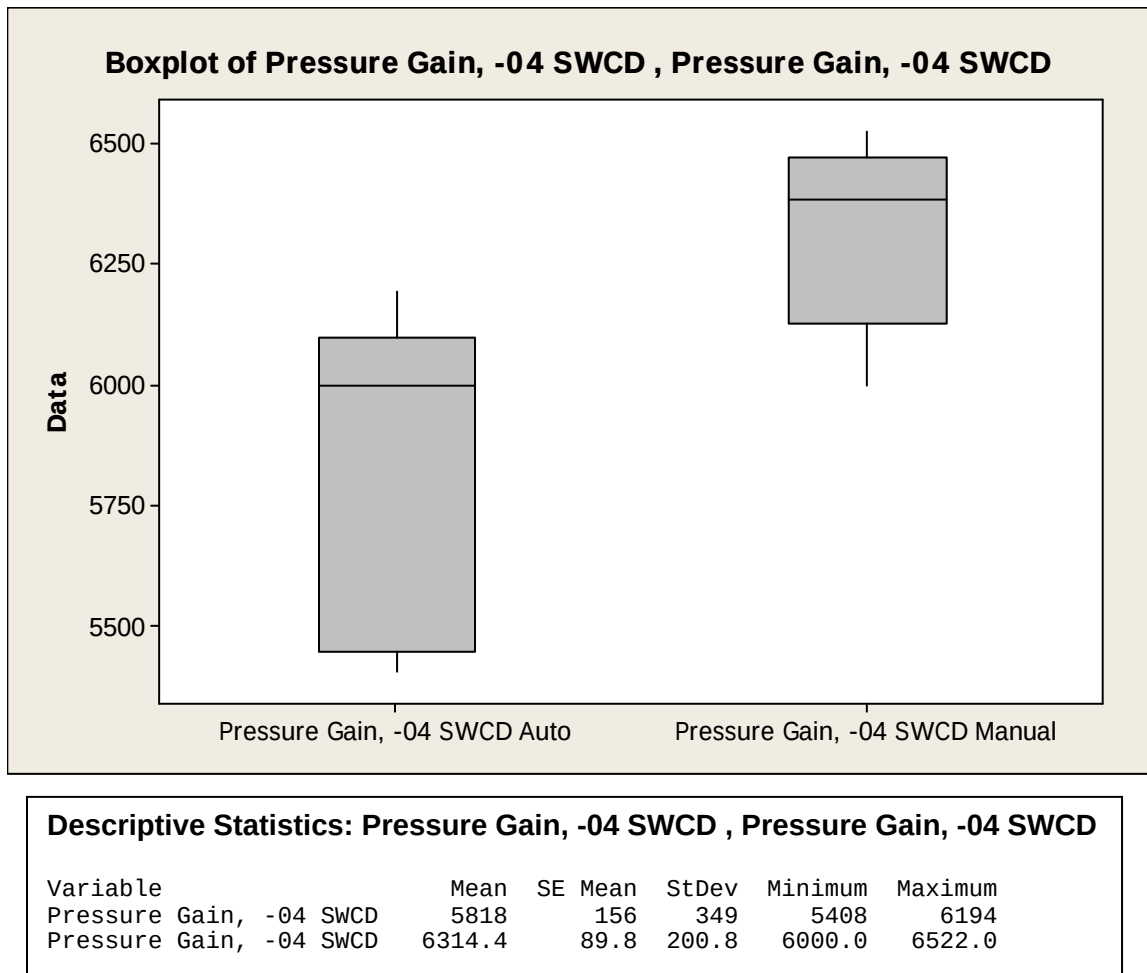


Figure 6. Boxplot of Pressure Gain for Servovalve P/N 22282790-104

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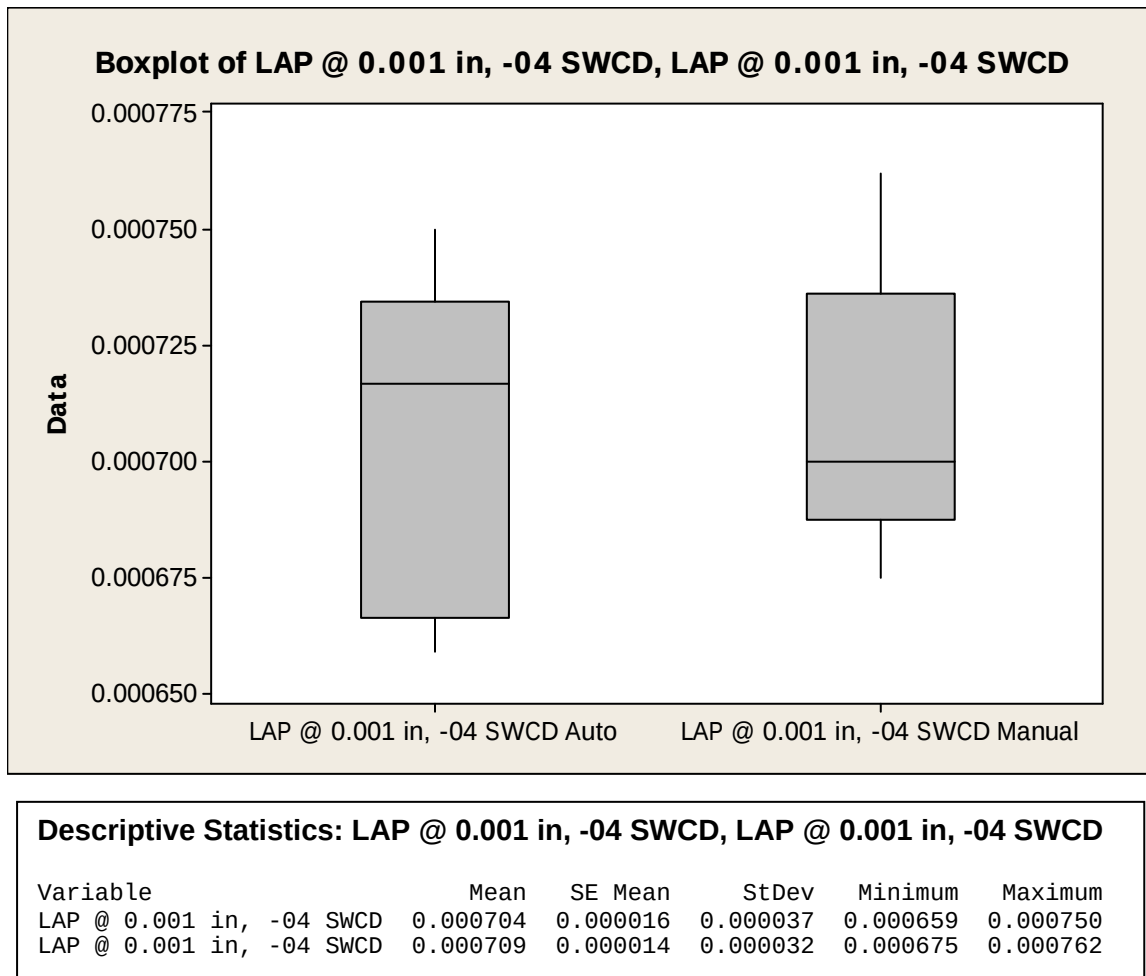


Figure 7. Boxplot of Lap at 0.001 in. for Servovalve P/N 22284090-104

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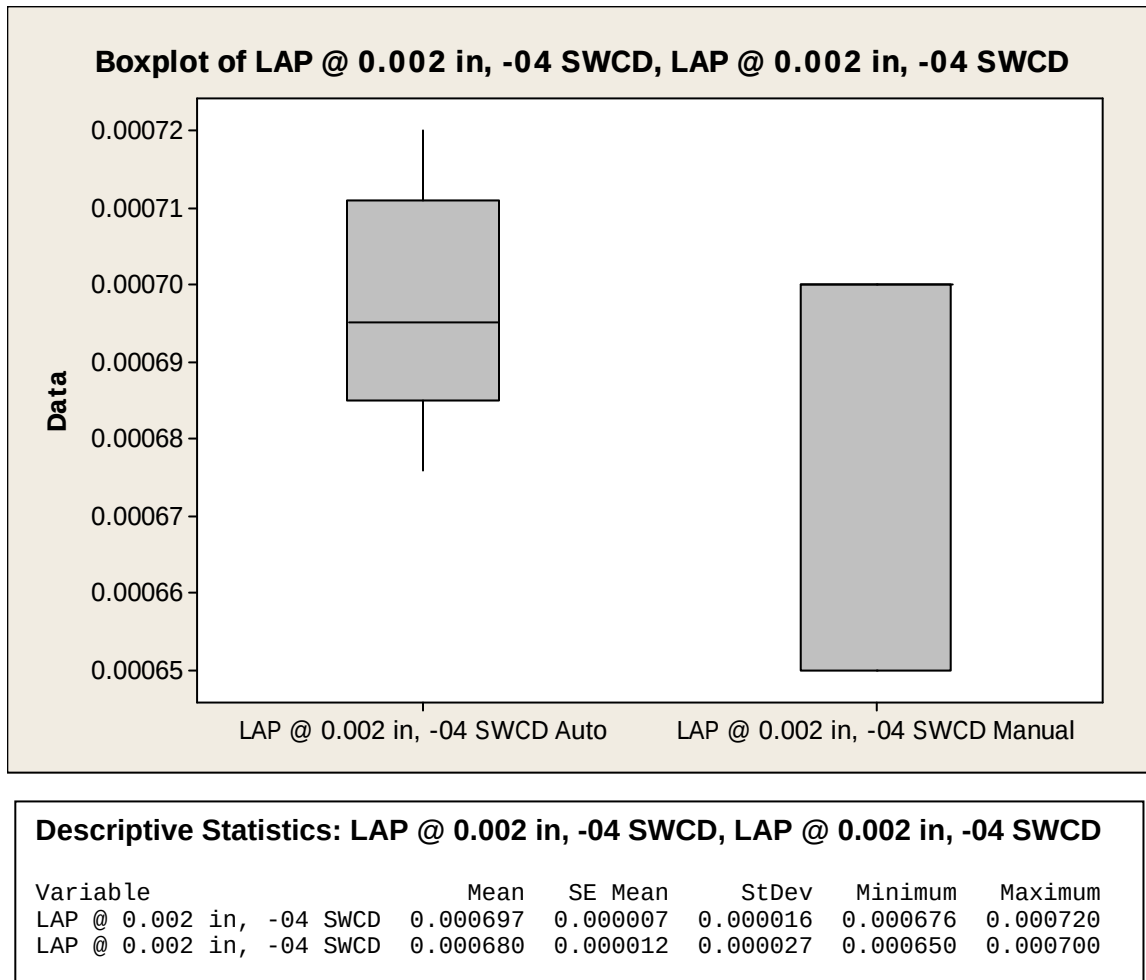


Figure 8. Boxplot of Lap at 0.002 in. for Servovalve P/N 22284090-104

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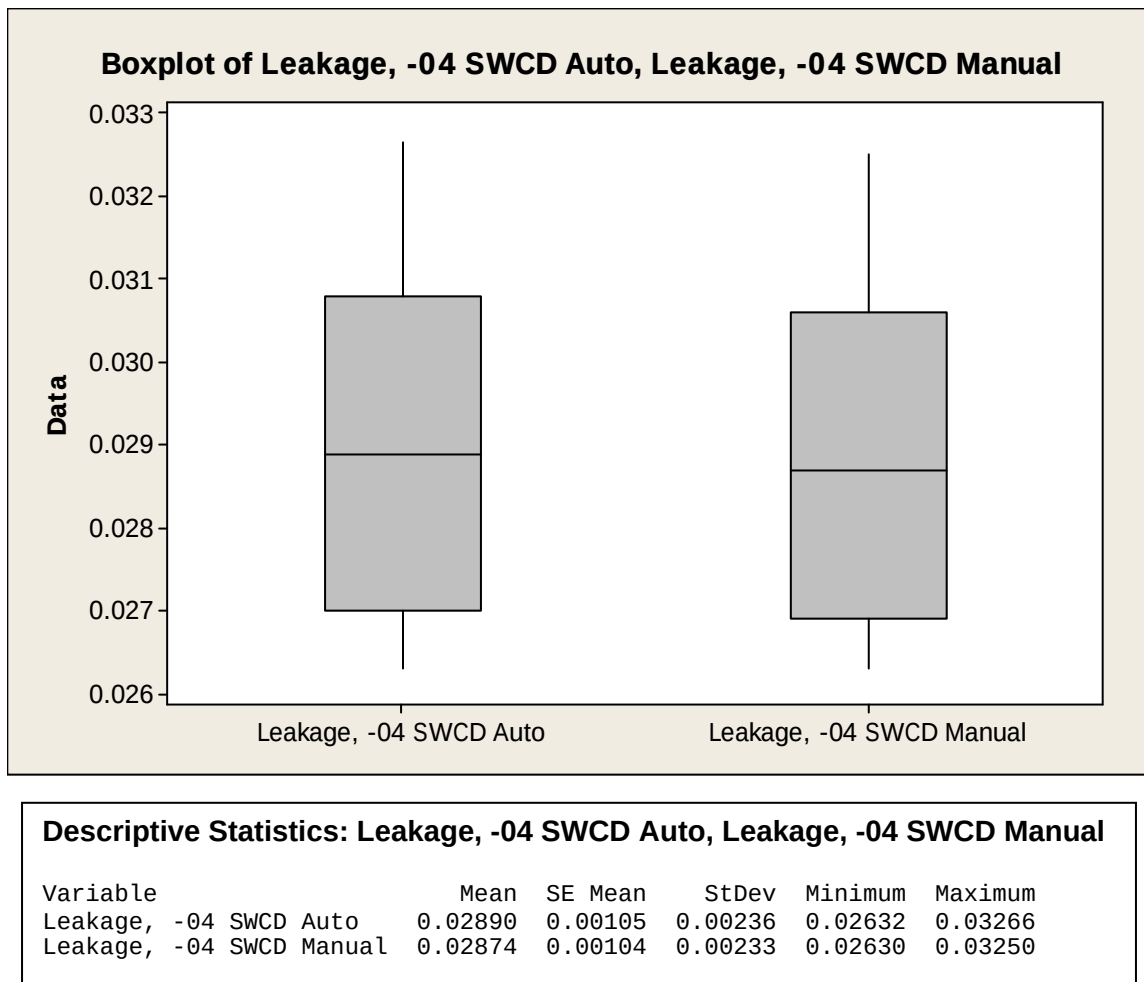


Figure 9. Boxplot of Leakage for Servovalve P/N 22284090-104

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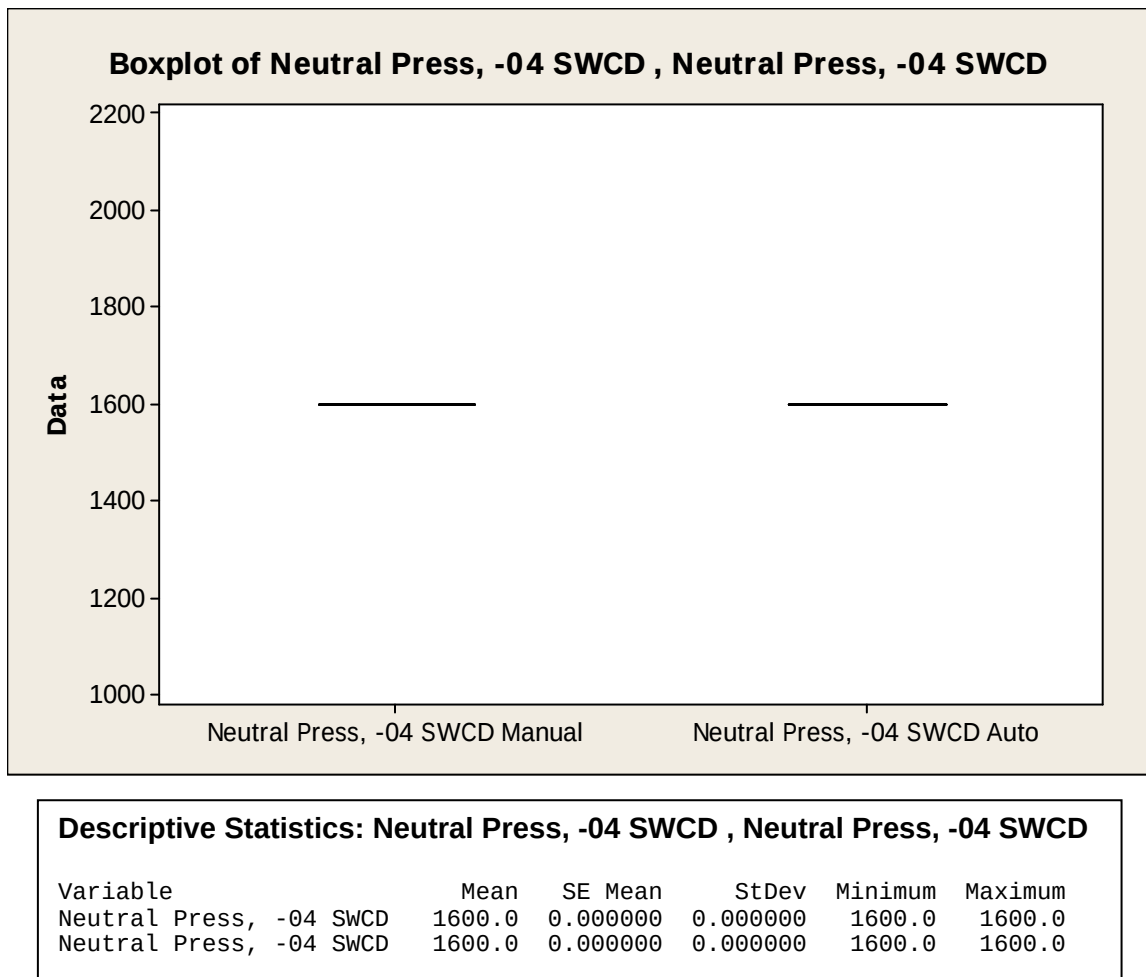


Figure 10. Boxplot of Neutral Press for Servovalve P/N 22284090-104

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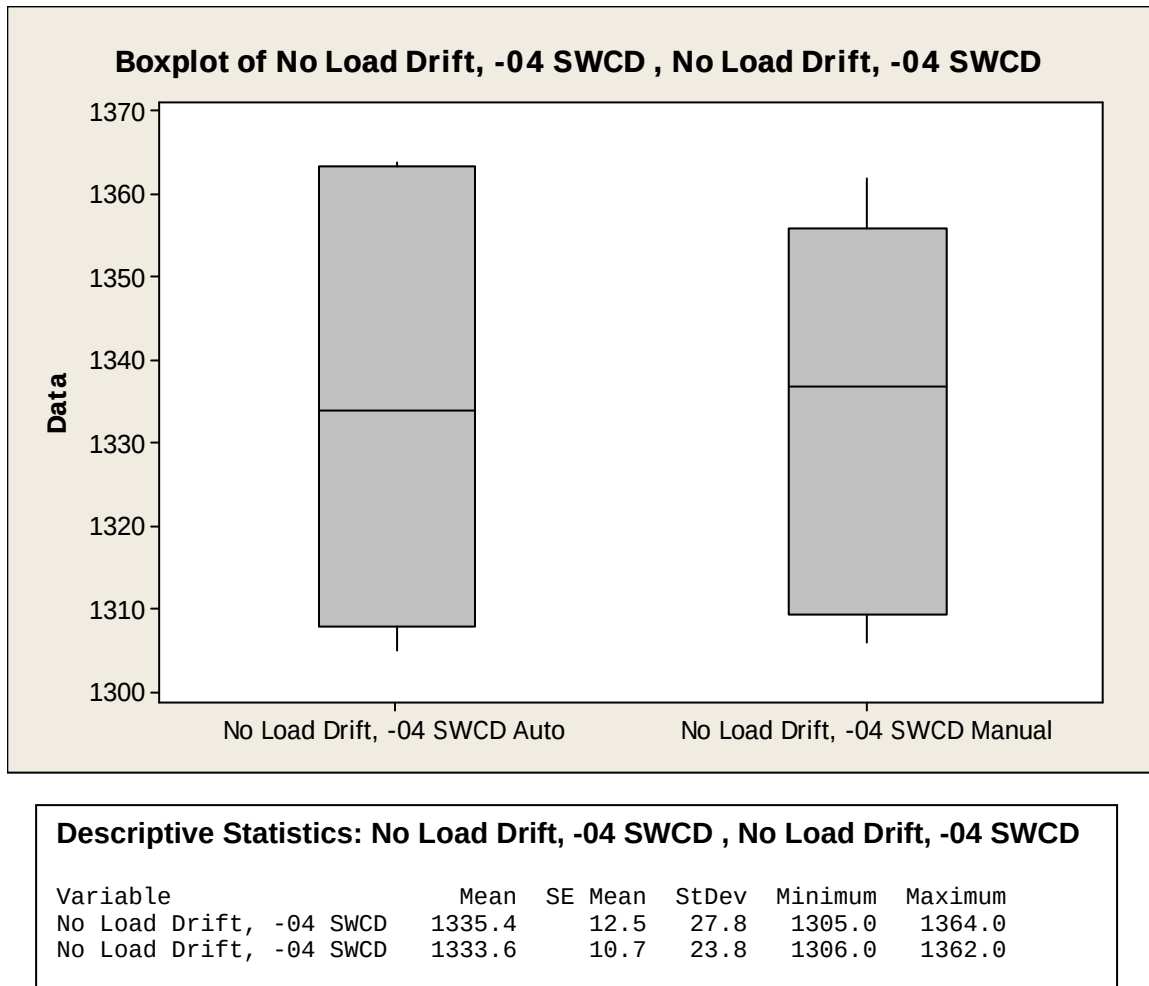
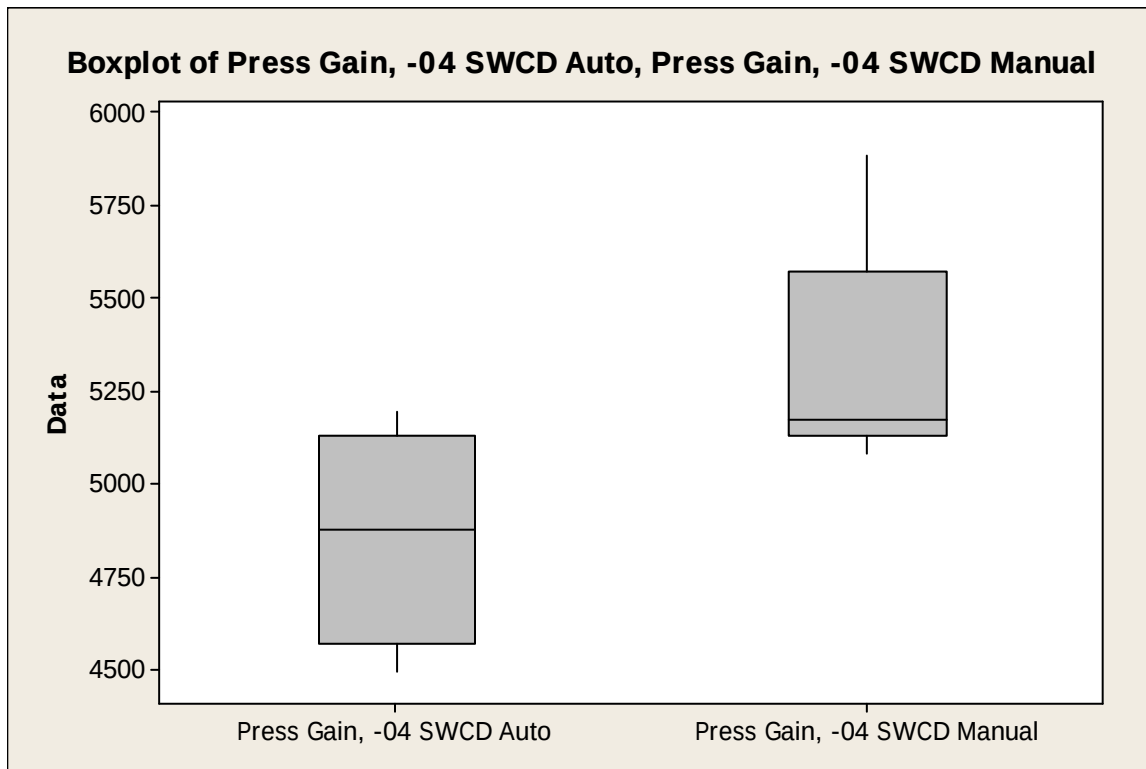


Figure 11. Boxplot of No-load Drift for Servovalve P/N 22284090-104

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Descriptive Statistics: Press Gain, -04 SWCD Auto, Press Gain, -04 SWCD Manual

Variable	Mean	SE Mean	StDev	Minimum	Maximum
Press Gain, -04 SWCD Aut	4856	129	288	4494	5195
Press Gain, -04 SWCD Man	5315	145	323	5084	5882

Figure 12. Boxplot of Pressure Gain for Servovalve P/N 22284090-104

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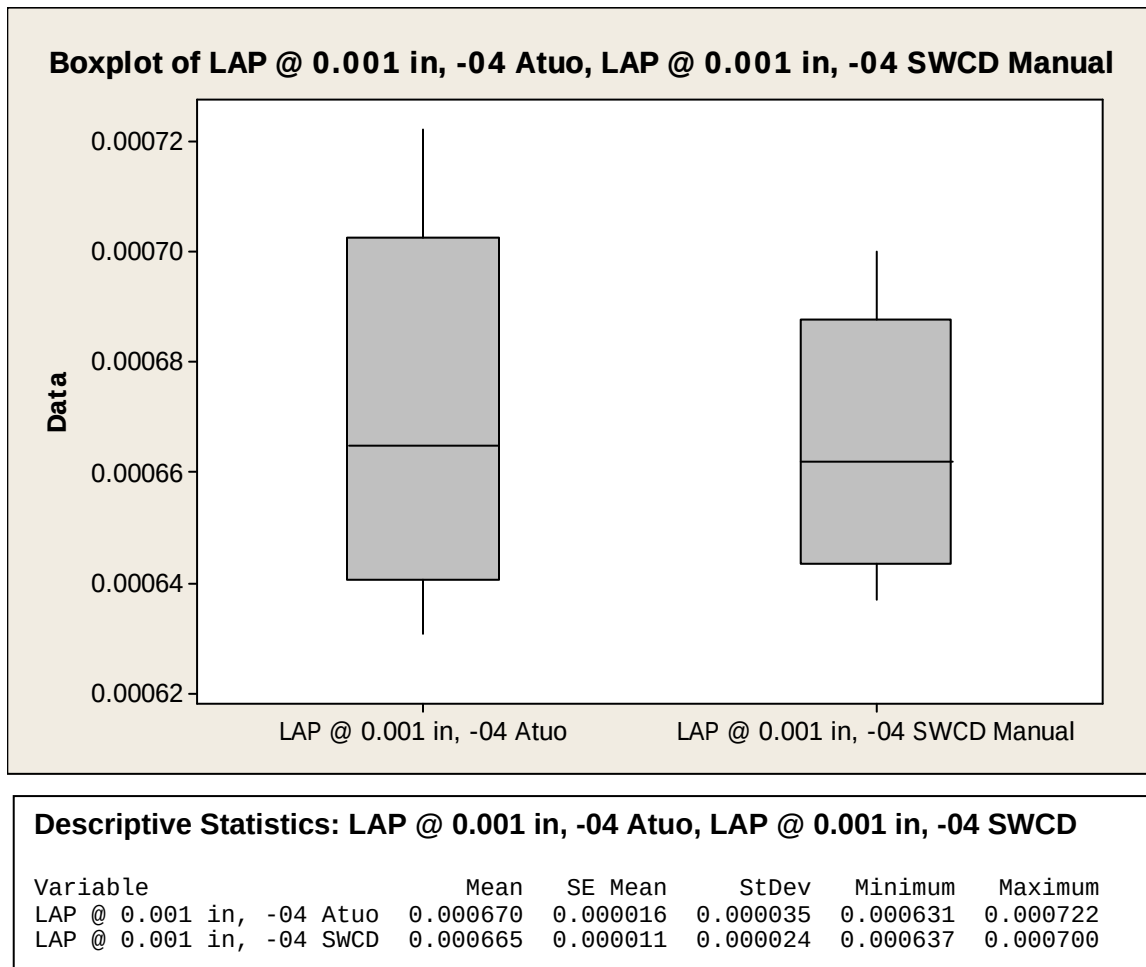


Figure 13. Boxplot of Lap at 0.001 in. for Servovalve P/N 22284250-104

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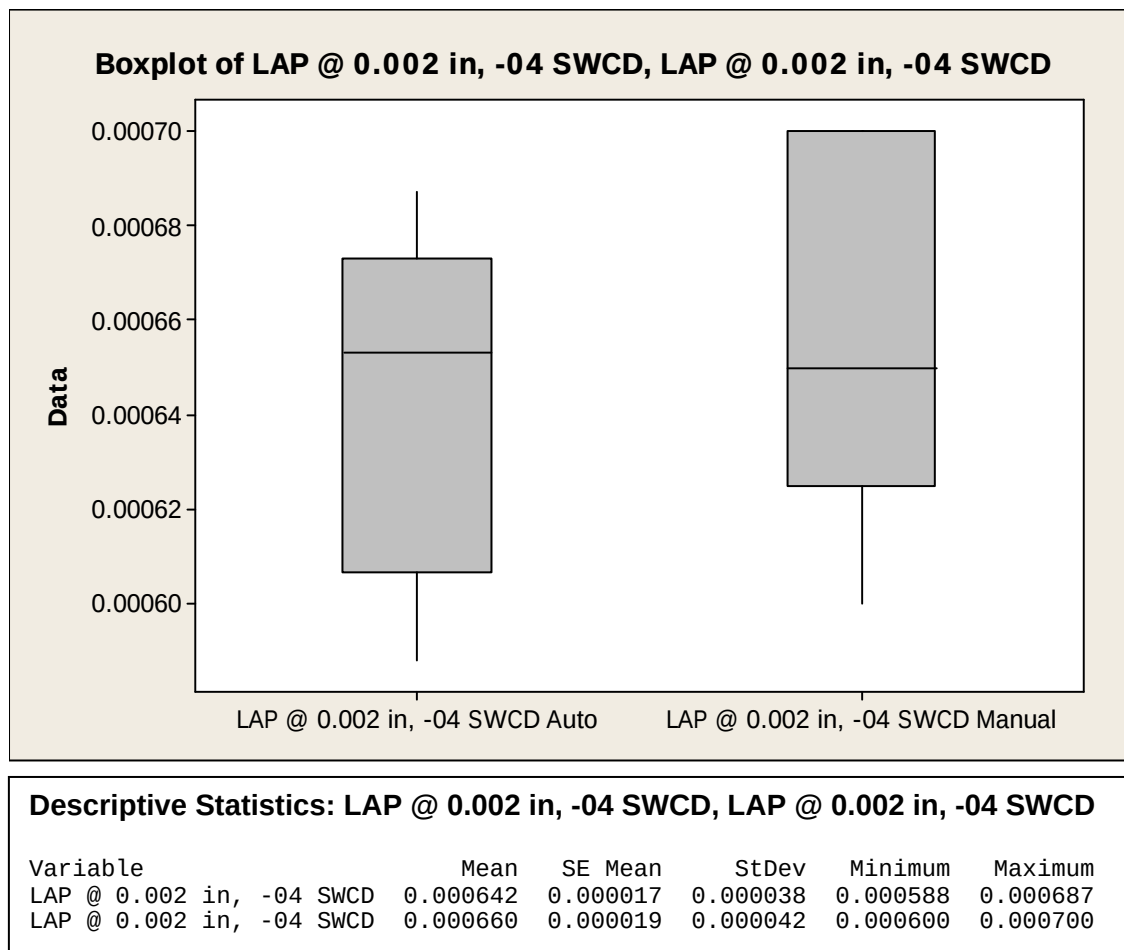


Figure 14. Boxplot of Lap at 0.002 in. for Servovalve P/N 22284250-104

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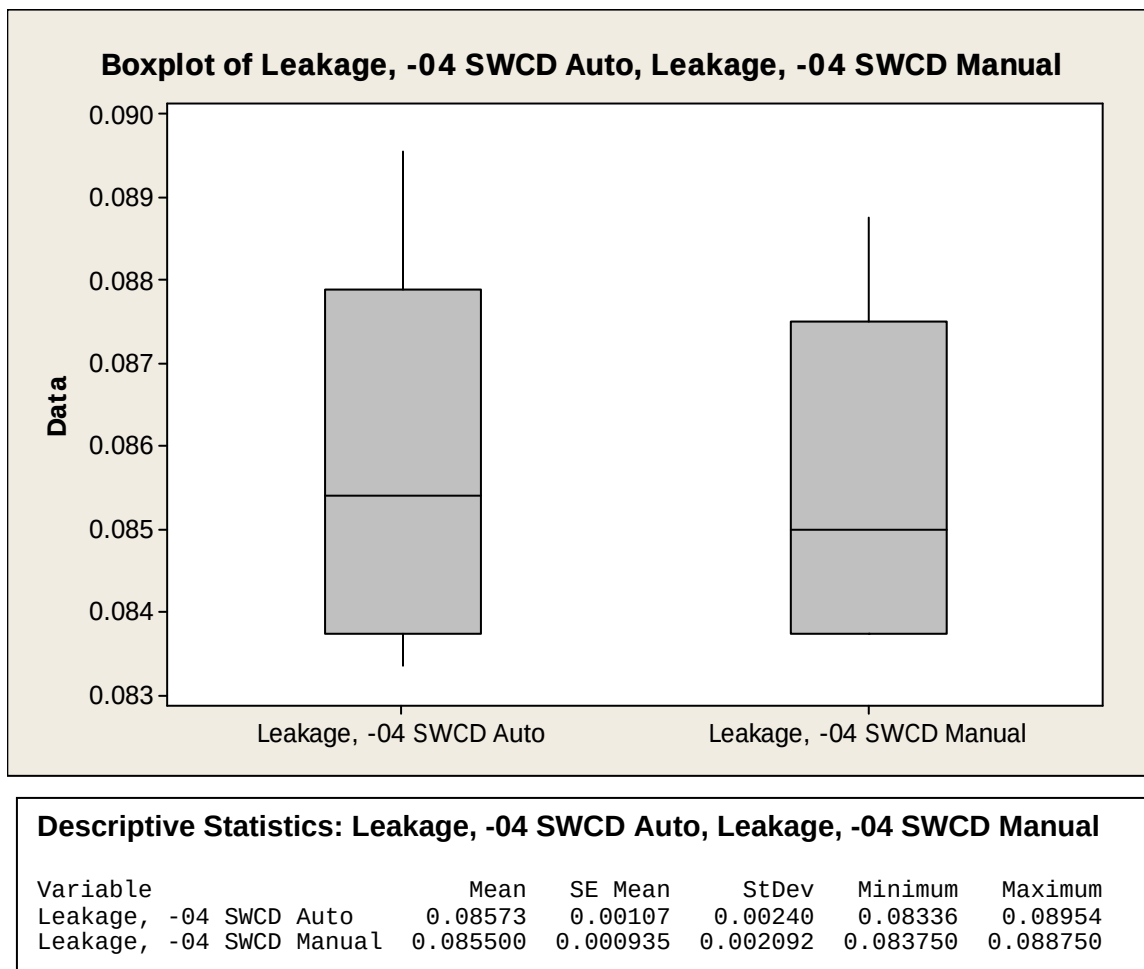


Figure 15. Boxplot of Leakage for Servovalve P/N 22284250-104

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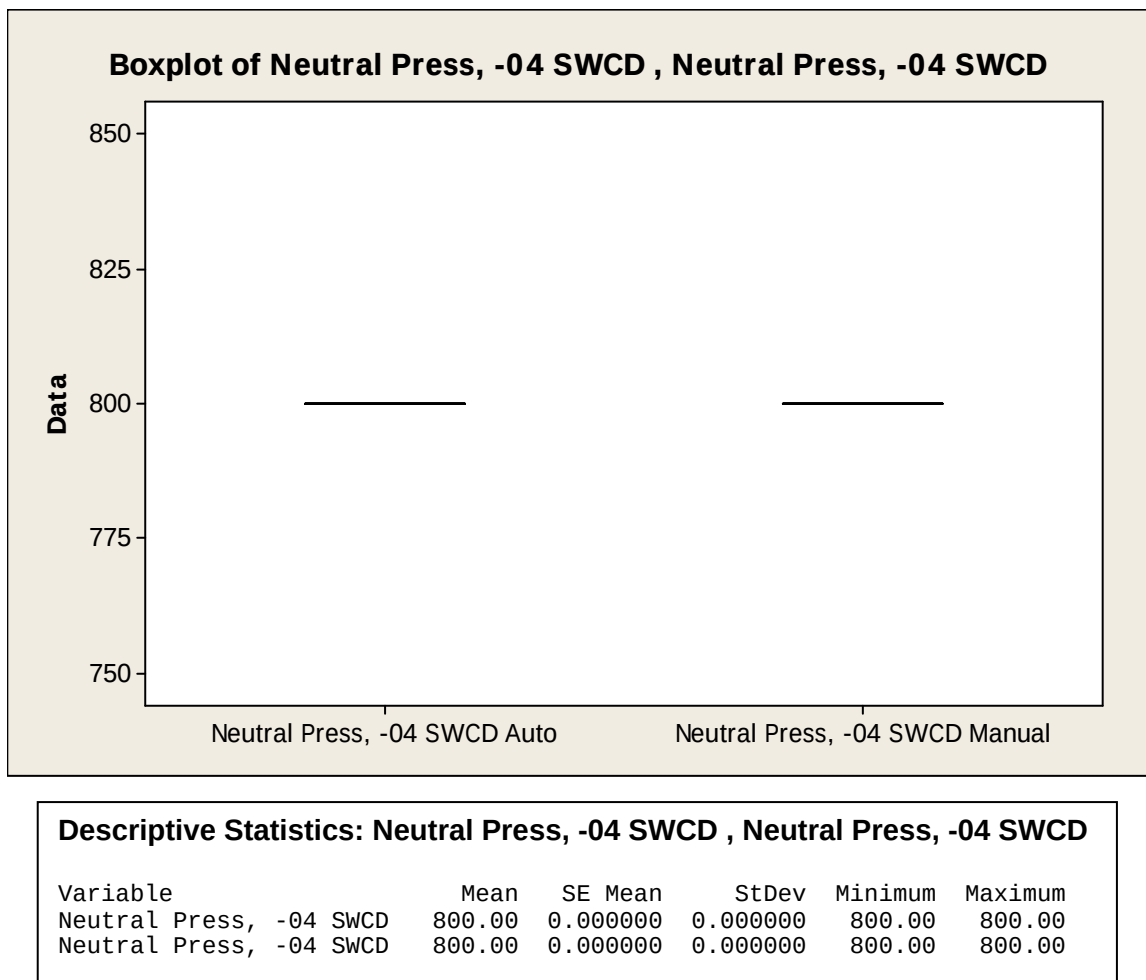


Figure 16. Boxplot of Neutral Pressure for Servovalve P/N 22284250-104

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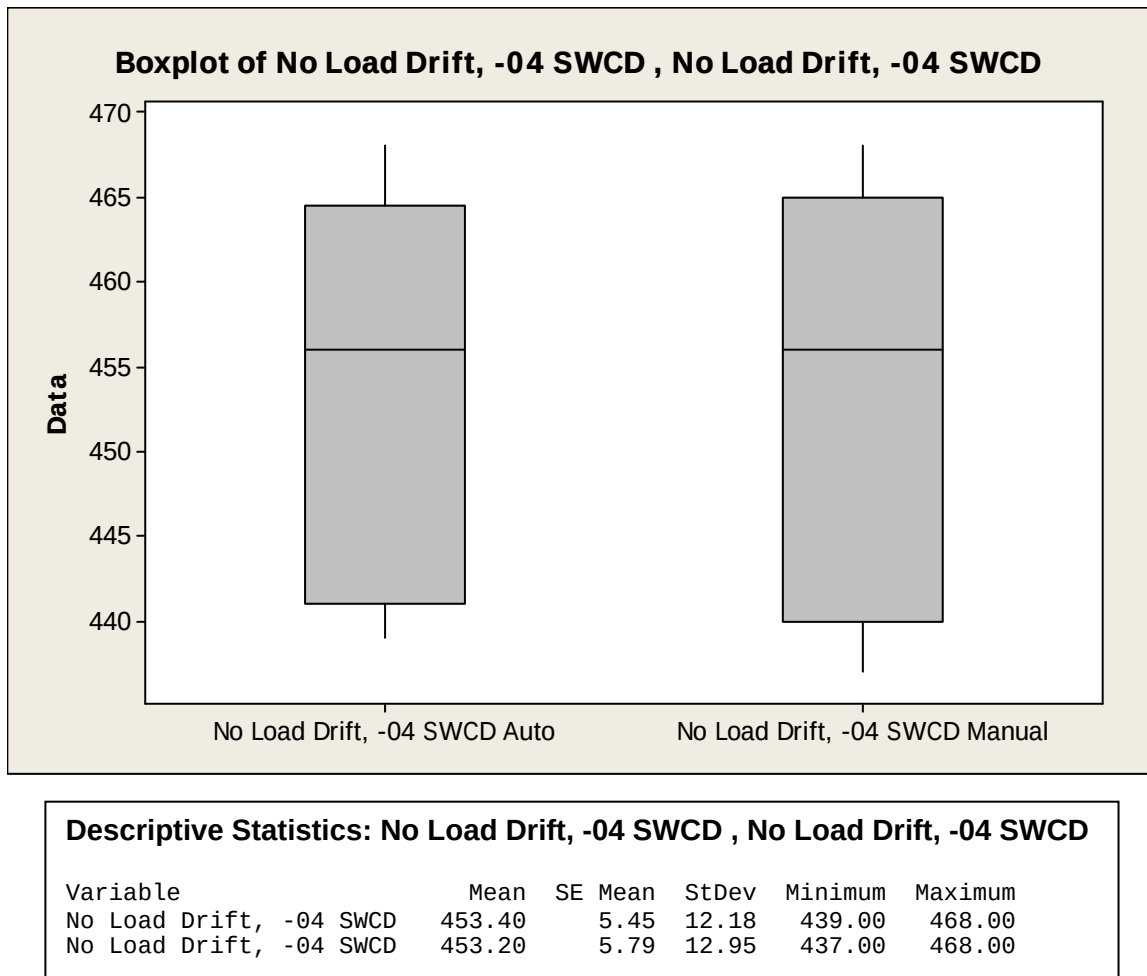


Figure 17. Boxplot of No-load Drift for Servovalve P/N 22284250-104

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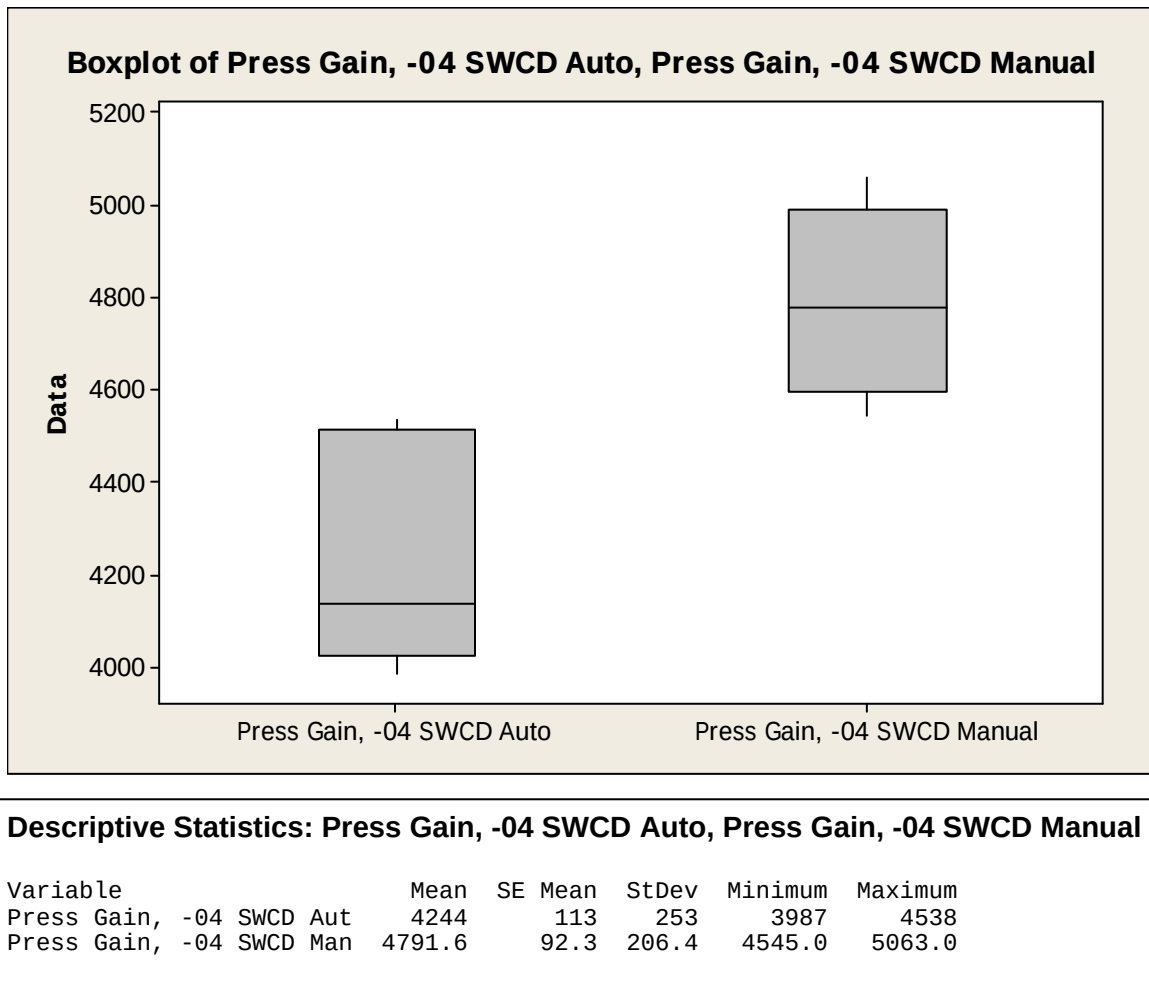


Figure 18. Boxplot of Pressure Gain for Servovalve P/N 22284250-104

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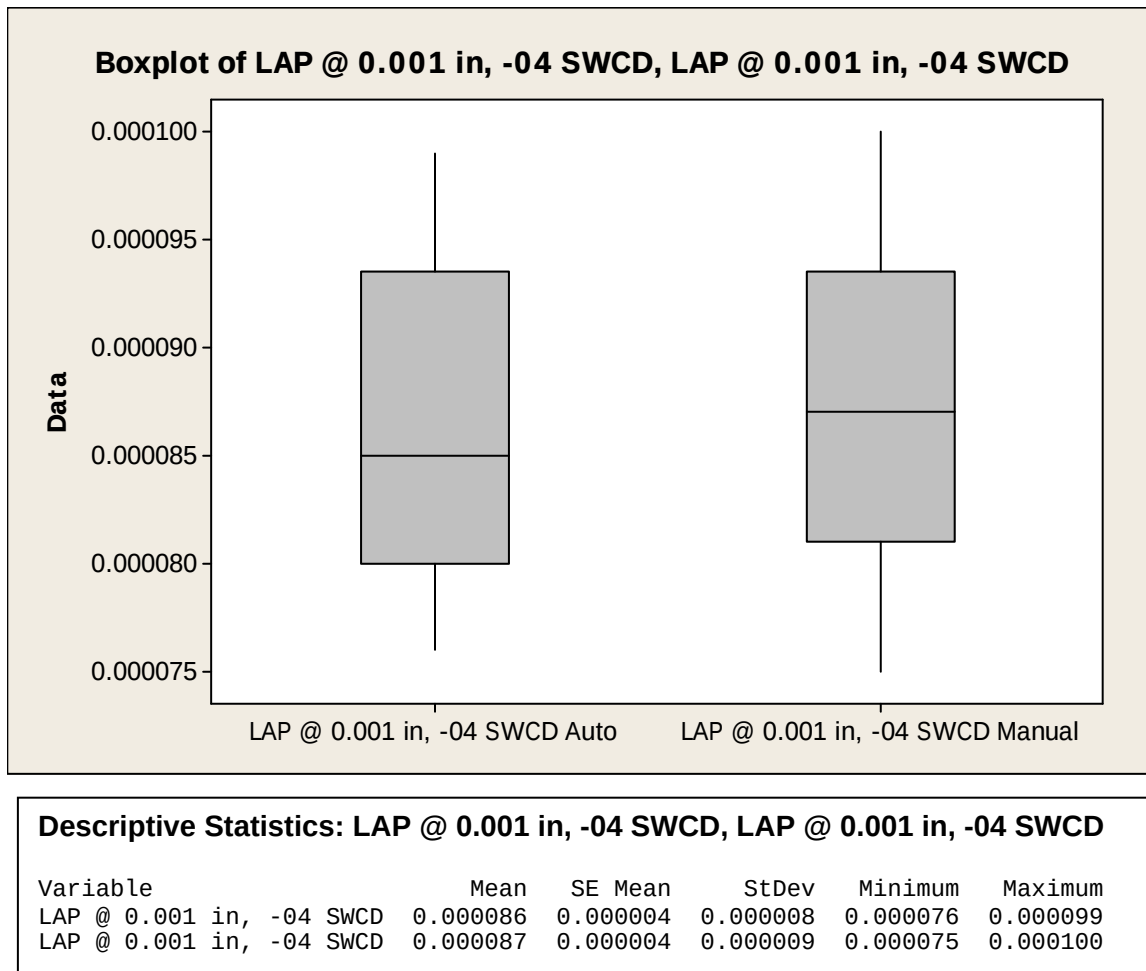


Figure 19. Boxplot of Lap at 0.001 in. for Servovalve P/N 22253726 (25A)

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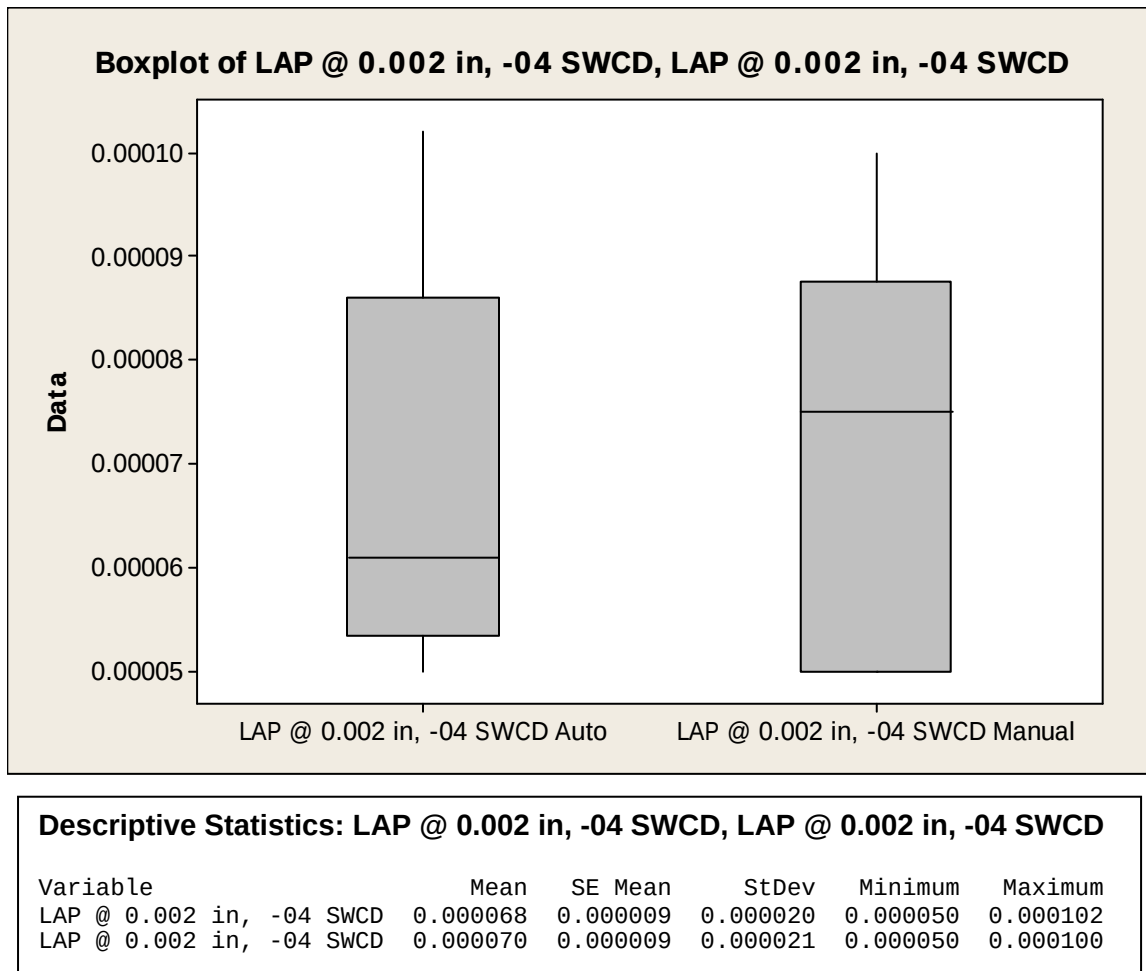


Figure 20. Boxplot of Lap at 0.002 in. for Servovalve P/N 22253726 (25A)

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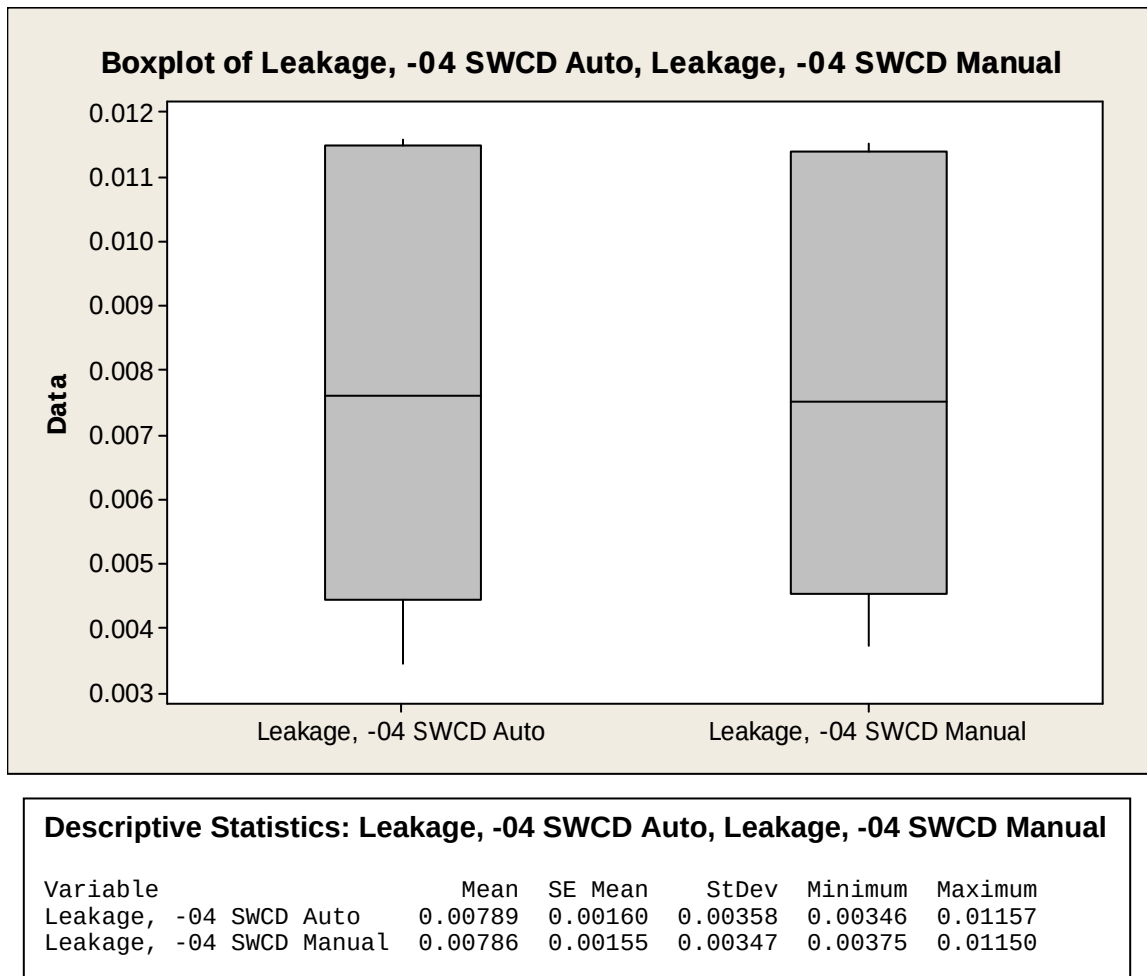
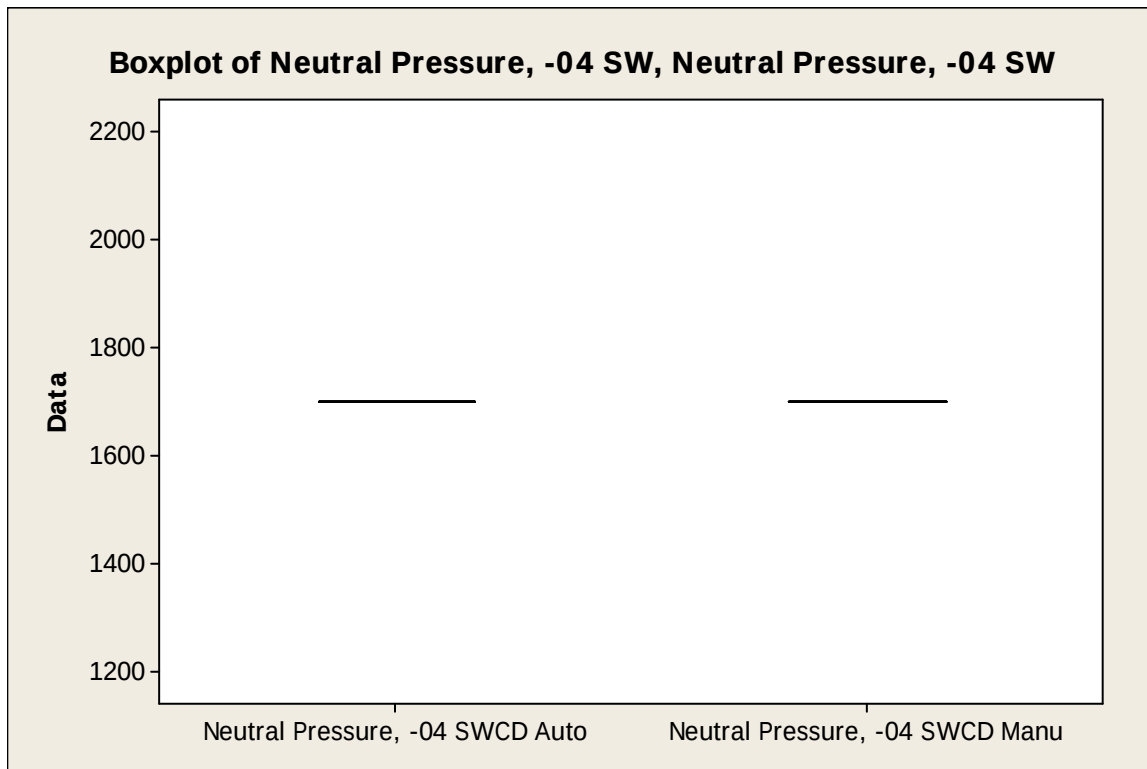


Figure 21. Boxplot of Leakage for Servovalve P/N 22253726 (25A)

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Descriptive Statistics: Neutral Pressure, -04 SW, Neutral Pressure, -04 SW

Variable	Mean	SE Mean	StDev	Minimum	Maximum
Neutral Pressure, -04 SW	1700.0	0.000000	0.000000	1700.0	1700.0
Neutral Pressure, -04 SW	1700.0	0.000000	0.000000	1700.0	1700.0

Figure 22. Boxplot of Neutral Pressure for Servovalve P/N 22253726 (25A)

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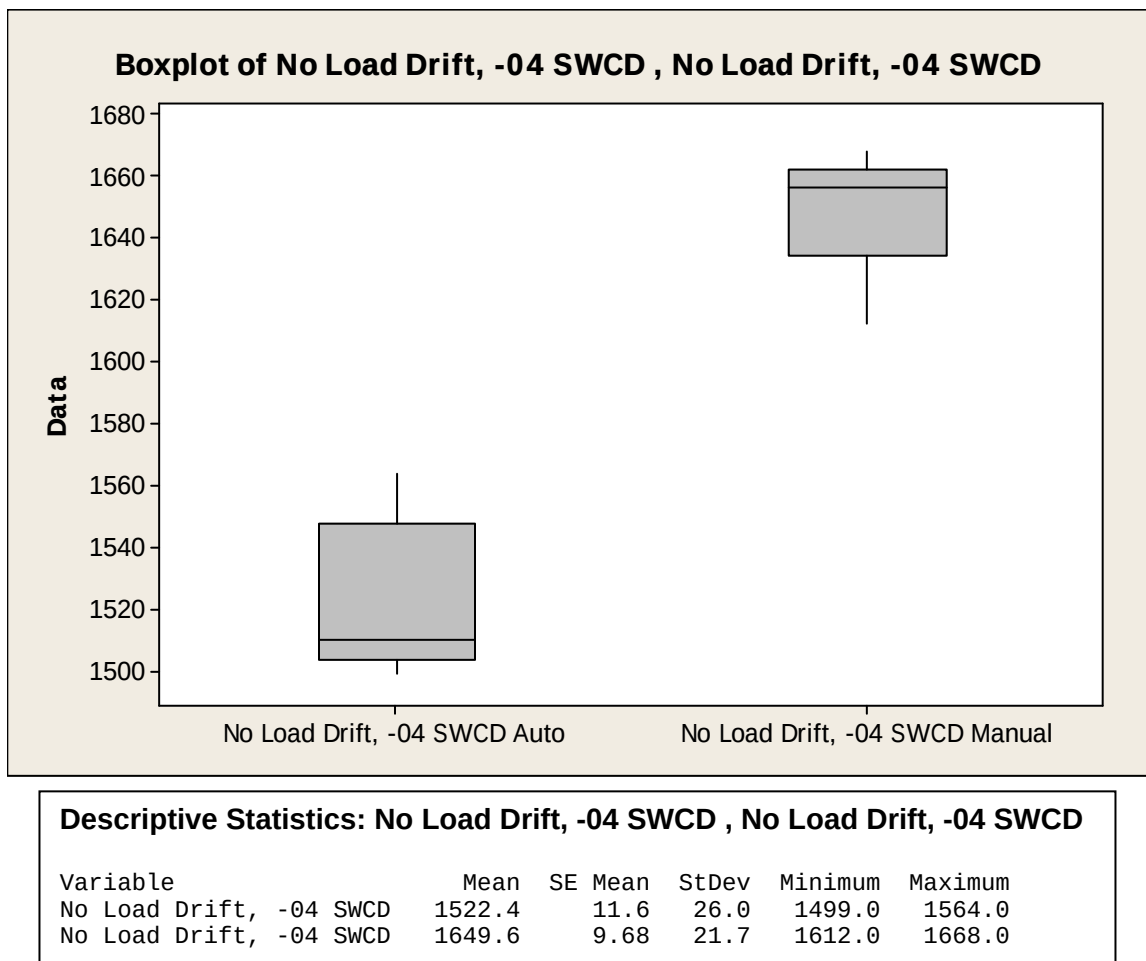


Figure 23. Boxplot of No-load Drift for Servovalve P/N 22253726 (25A)

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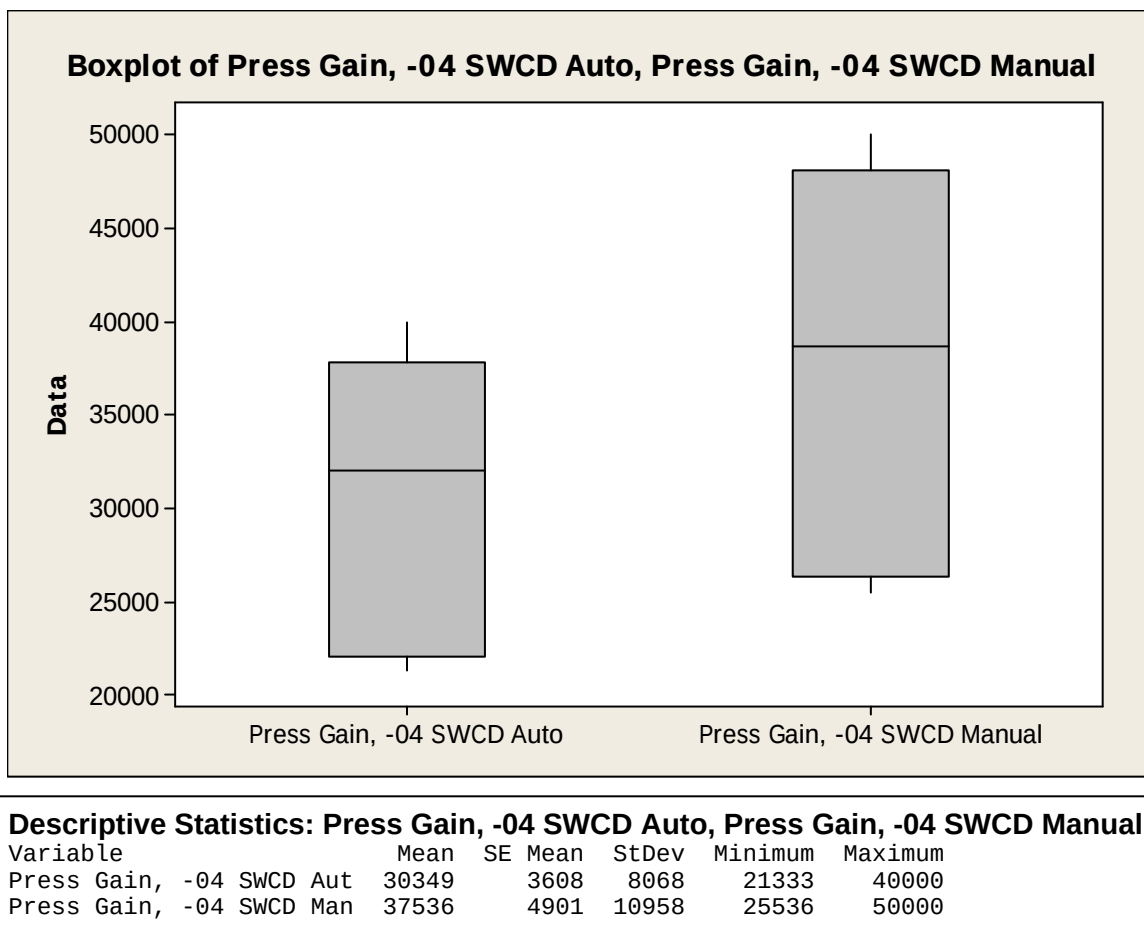


Figure 24. Boxplot of Pressure Gain for Servovalve P/N 22253726 (25A)

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Table 1. Test Result Analysis for P/N 22282790-104

Figure No.	Test	Lower Spec Limit	Upper Spec Limit	SWCD -04 Data Automated Counting Mean StDev	SWCD -04 Data Manual Counting Mean StDev	%Mean Diff with Respect to Spec. Limits	StDev Difference
1	Lap Condition at 0.001/in.	0.0003	0.0004	0.000315 0.000010	0.000312 0.000013	3.0	-0.000003
2	Lap Condition at 0.002/in.	0.0003	0.0004	0.000333 0.000043	0.000324 0.000043	9.0	0.0
3	Leakage (gpm)	-	0.3000	0.20123 0.00207	0.20072 0.00192	0.2	0.000150
4	Neutral Pressure (psi)	800.00	2200.00	1300.0 0.000000	1300.0 0.000000	0.0	0.0
5	No-load Drift (psi)	-	-	1247.4 27.9	1247.6 28.9	N/A	-1.0
6	Pressure Gain (psi/0.001 in.) at 1200 psi	4600.00	-	5818 349	6314.4 200.8	-10.8	148.2

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Table 2. Test Result Analysis for P/N 22284090-104

Figure No.	Test	Lower Spec Limit	Upper Spec Limit	SWCD -04 Data Automated Counting Mean StDev	SWCD -04 Data Manual Counting Mean StDev	%Mean Diff with Respect to Spec. Limits	StDev Difference
7	Lap Condition at 0.001/in.	0.0008	0.0010	0.000704 0.000037	0.000709 0.000032	-2.5	0.000005
8	Lap Condition at 0.002/in.	0.0008	0.0010	0.000697 0.000016	0.000680 0.000027	8.5	-0.000011
9	Leakage (gpm)	—	0.0750	0.02890 0.00236	0.02874 0.00233	0.2	0.000030
10	Neutral Pressure (psi)	600.00	2400.00	1600.0 0.000000	1600.0 0.000000	0.0	0.0
11	No-load Drift (psi)	—	—	1335.4 27.8	1333.6 23.8	N/A	4.0
12	Pressure Gain (psi/0.001 in.) at 1200 psi	2700.00	—	4856 288	5315 323	-17.0	-35.0

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Table 3. Test Result Analysis for P/N 22284250-104

Figure No.	Test	Lower Spec Limit	Upper Spec Limit	SWCD -04 Data Automated Counting Mean StDev	SWCD -04 Data Manual Counting Mean StDev	%Mean Diff with Respect to Spec. Limits	StDev Difference
13	Lap Condition at 0.001/in.	0.0004	0.0008	0.000670 0.000035	0.000665 0.000024	1.3	0.000011
14	Lap Condition at 0.002/in.	0.0004	0.0008	0.000642 0.000038	0.000660 0.000042	-4.5	-0.000004
15	Leakage (gpm)	—	0.0900	0.08573 0.00240	0.085500 0.002092	0.3	0.000308
16	Neutral Pressure (psi)	700.00	1000.00	800.00 0.000000	800.00 0.000000	0.0	0.0
17	No-load Drift (psi)	—	2900.00	453.40 12.18	453.20 12.95	0.0	-0.77
18	Pressure Gain (psi/0.001 in.) at 1200 psi	3500.00	—	4244 253	4791.6 206.4	-15.6	46.6

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Table 4. Test Result Analysis for P/N 22253726 (25A)

Figure No.	Test	Lower Spec Limit	Upper Spec Limit	SWCD -04 Data Automated Counting Mean StDev	SWCD -04 Data Manual Counting Mean StDev	%Mean Diff with Respect to Spec. Limits	StDev Difference
19	Lap Condition at 0.001/in.	-0.0001	0.0002	0.000086 0.000008	0.000087 0.000009	-0.3	-0.000001
20	Lap Condition at 0.002/in.	-0.0001	0.0002	0.000068 0.000020	0.000070 0.000021	-0.7	-0.000001
21	Leakage (gpm)	-	0.0300	0.00789 0.00358	0.00786 0.00347	0.1	0.000110
22	Neutral Pressure (psi)	1000.00	2000.00	1700.0 0.000000	1700.0 0.000000	0.0	0.0
23	No-load Drift (psi)	-	-	1522.4 26.0	1649.6 21.7	N/A	4.3
24	Pressure Gain (psi/0.001 in) at 1200 psi	7300	-	30349 8068	37536 10958	-98.5	-2890.0

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APPENDIX A

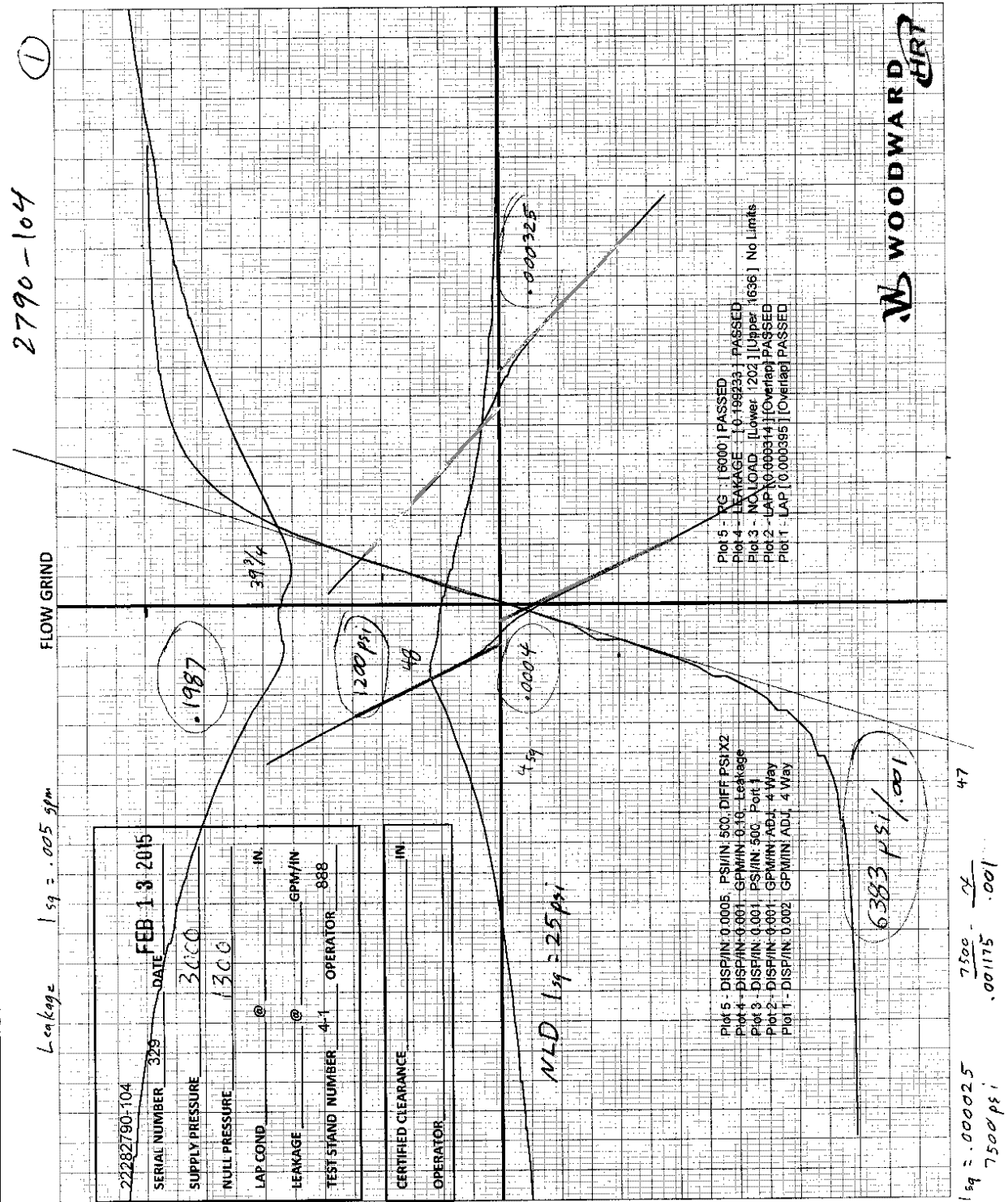
DATA PLOTS FOR SWCD 97900503 VERSION -04

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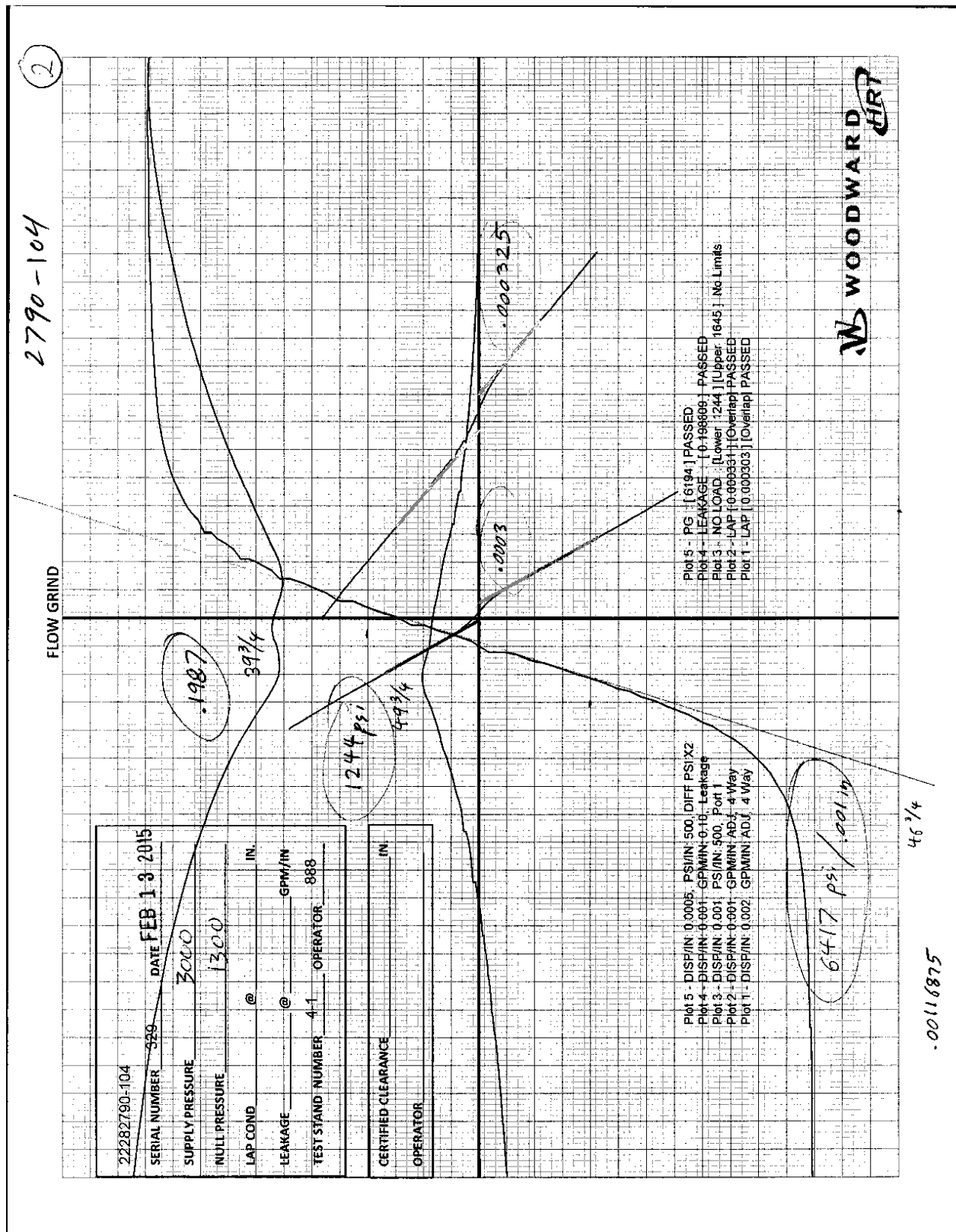
Appendix A-1

Servovalve 22282790-104

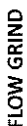
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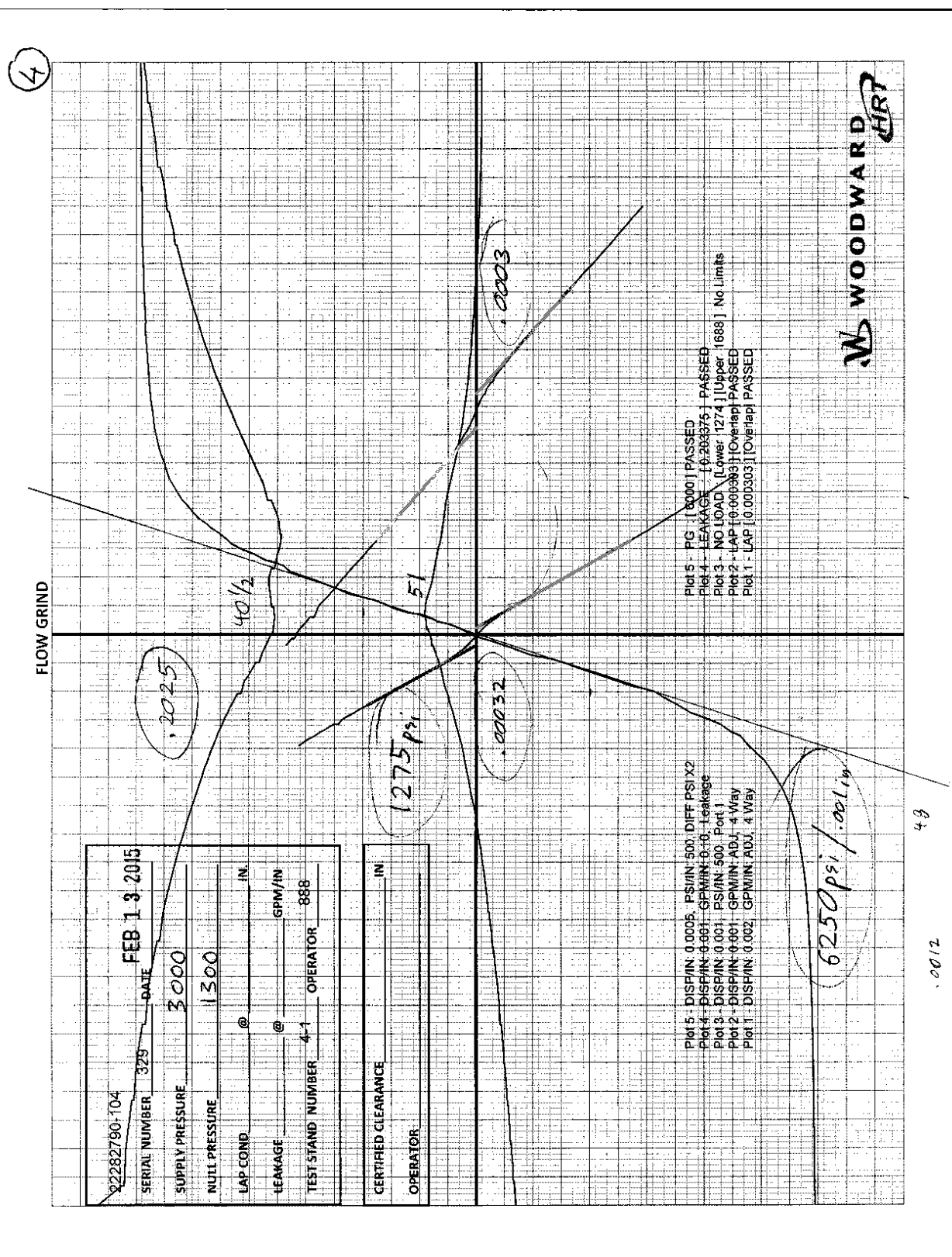
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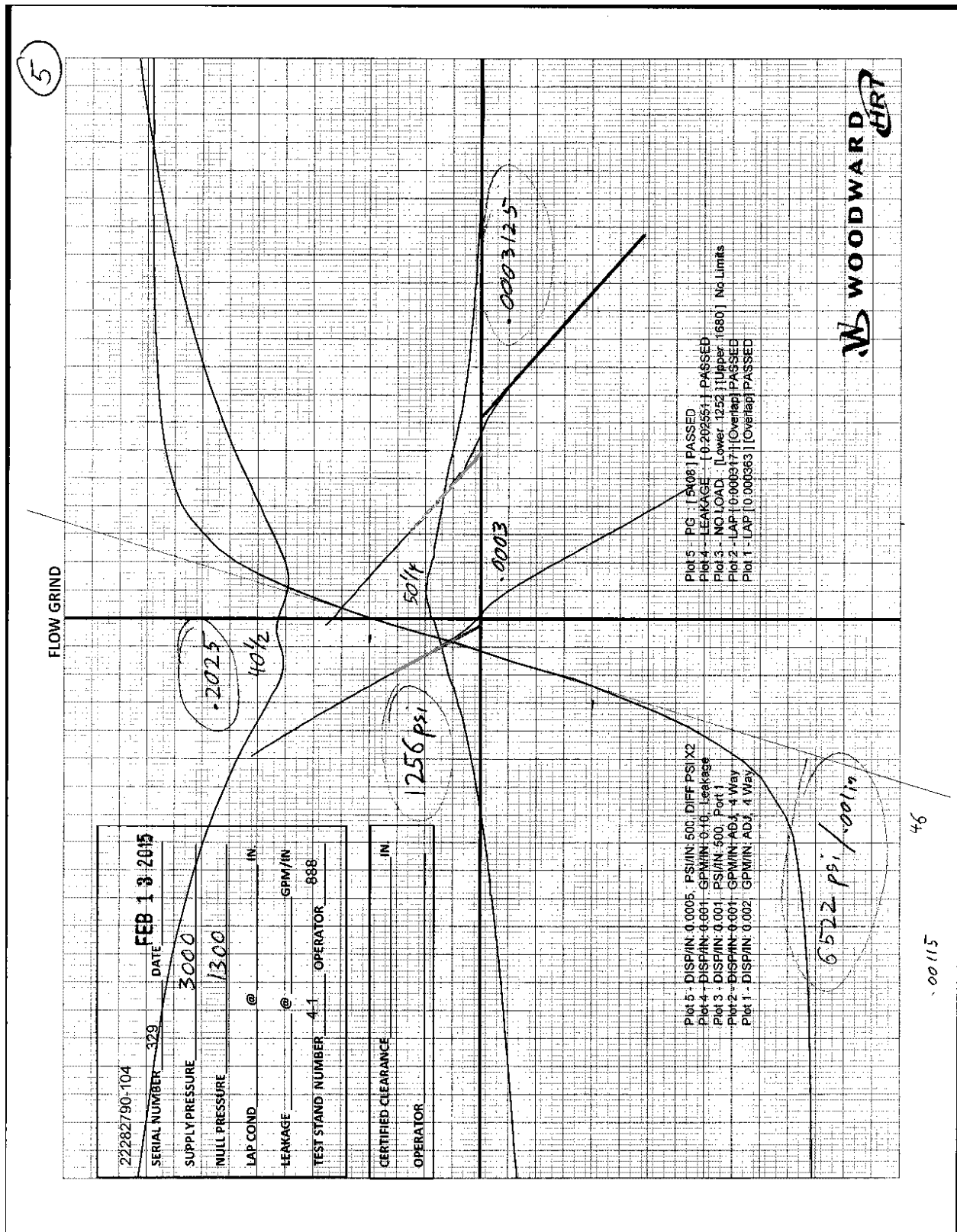


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PAGE NO.
A-5
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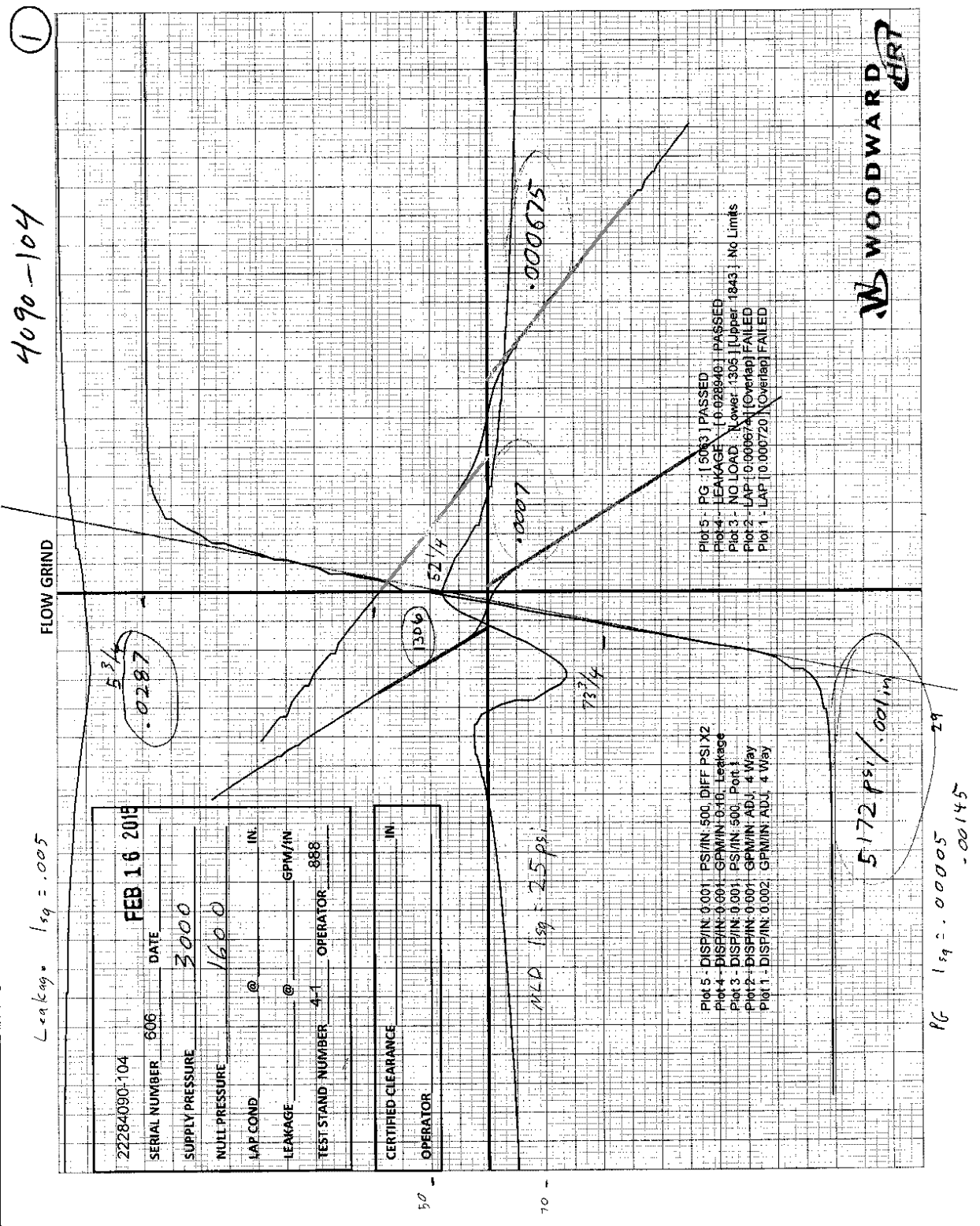


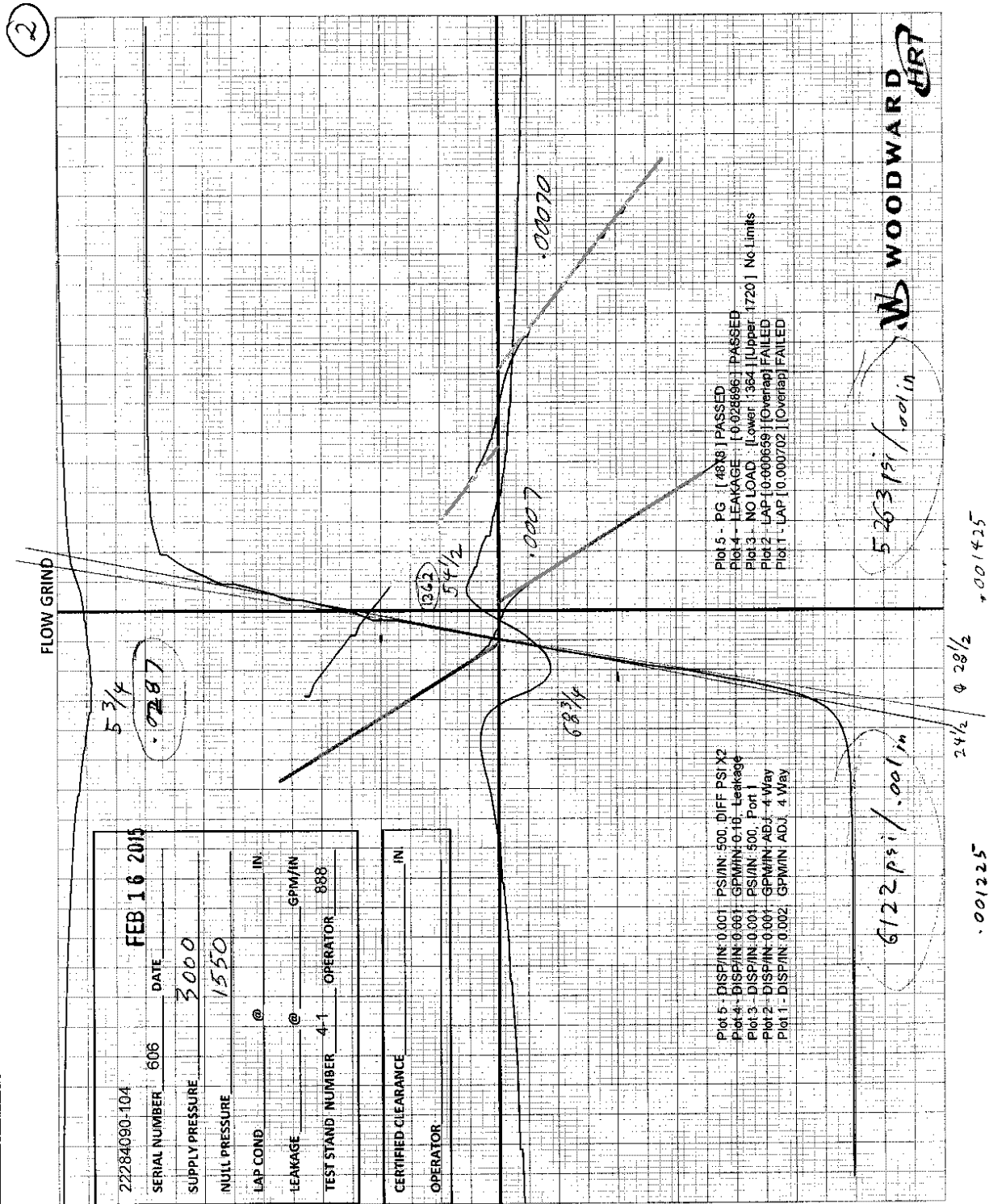
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Appendix A-2

Servovalve 22284090-104

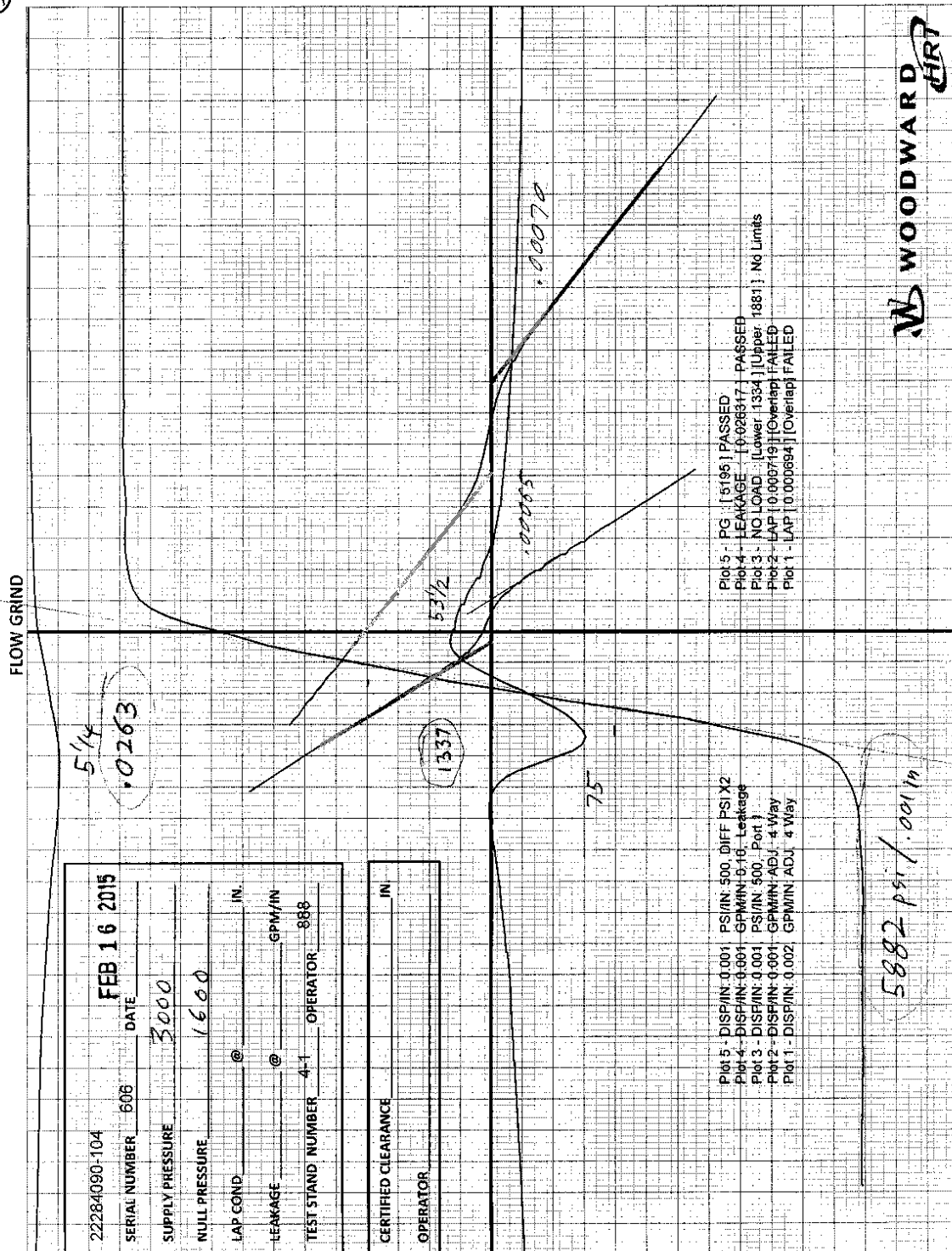
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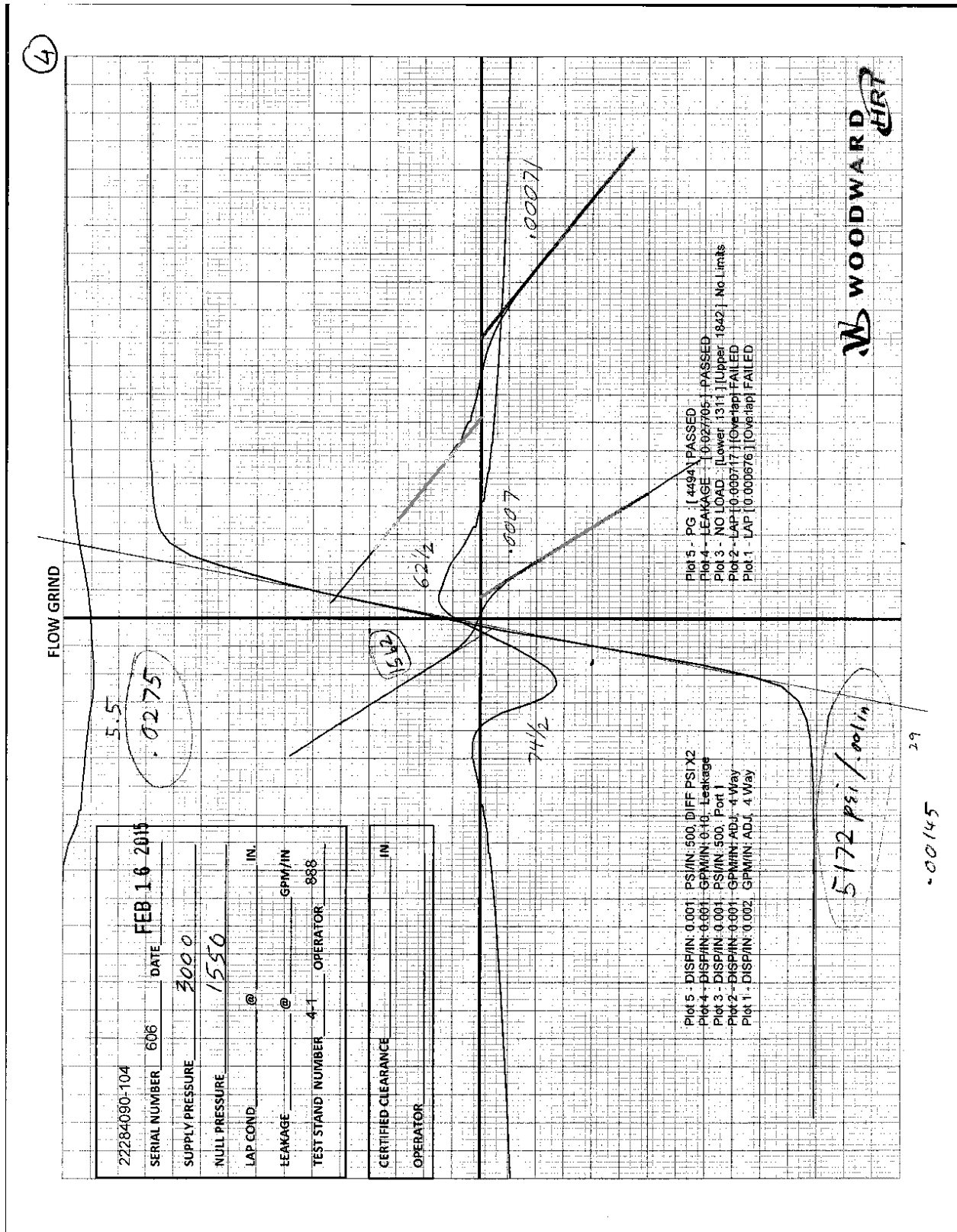


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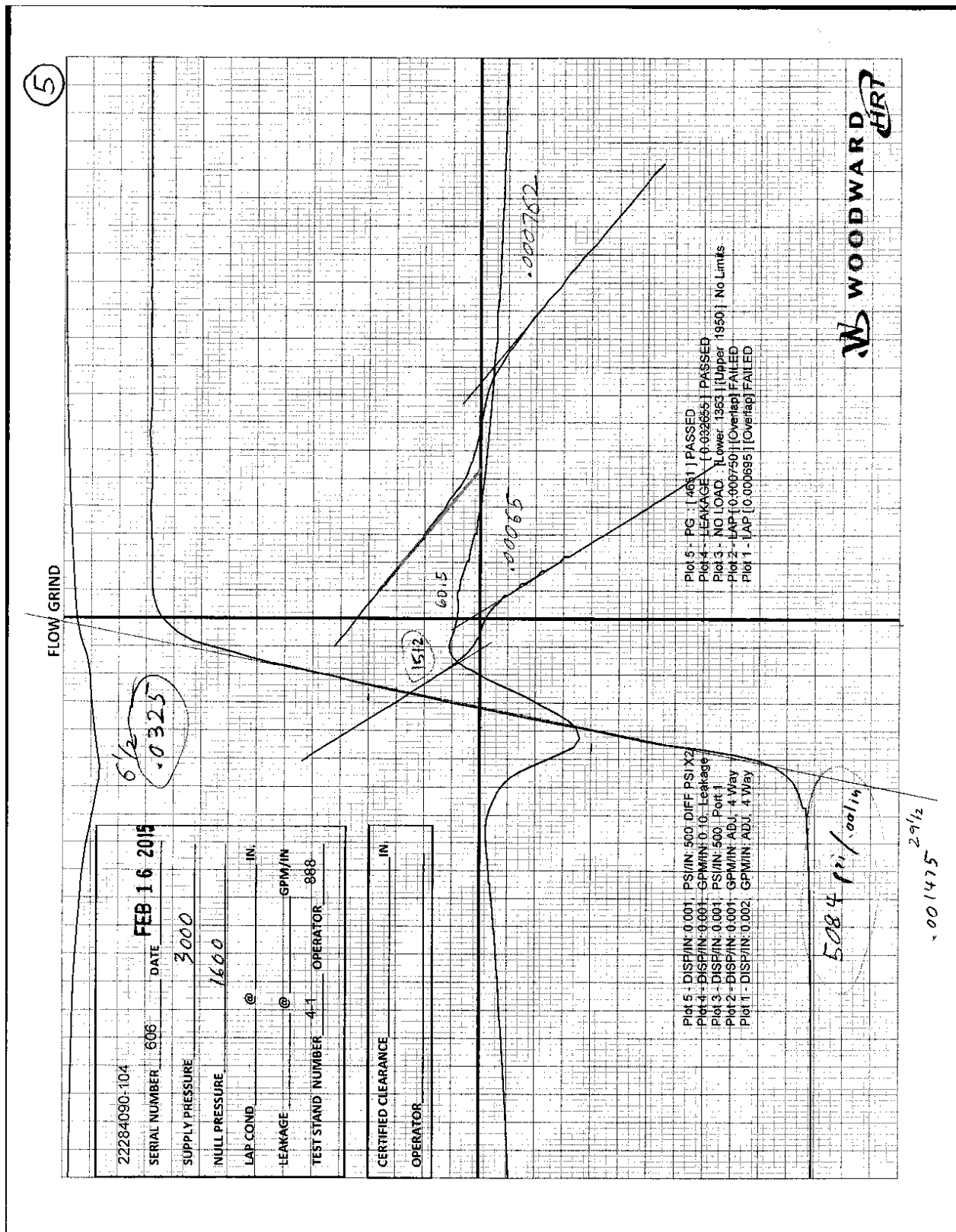
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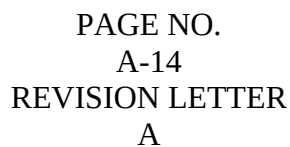


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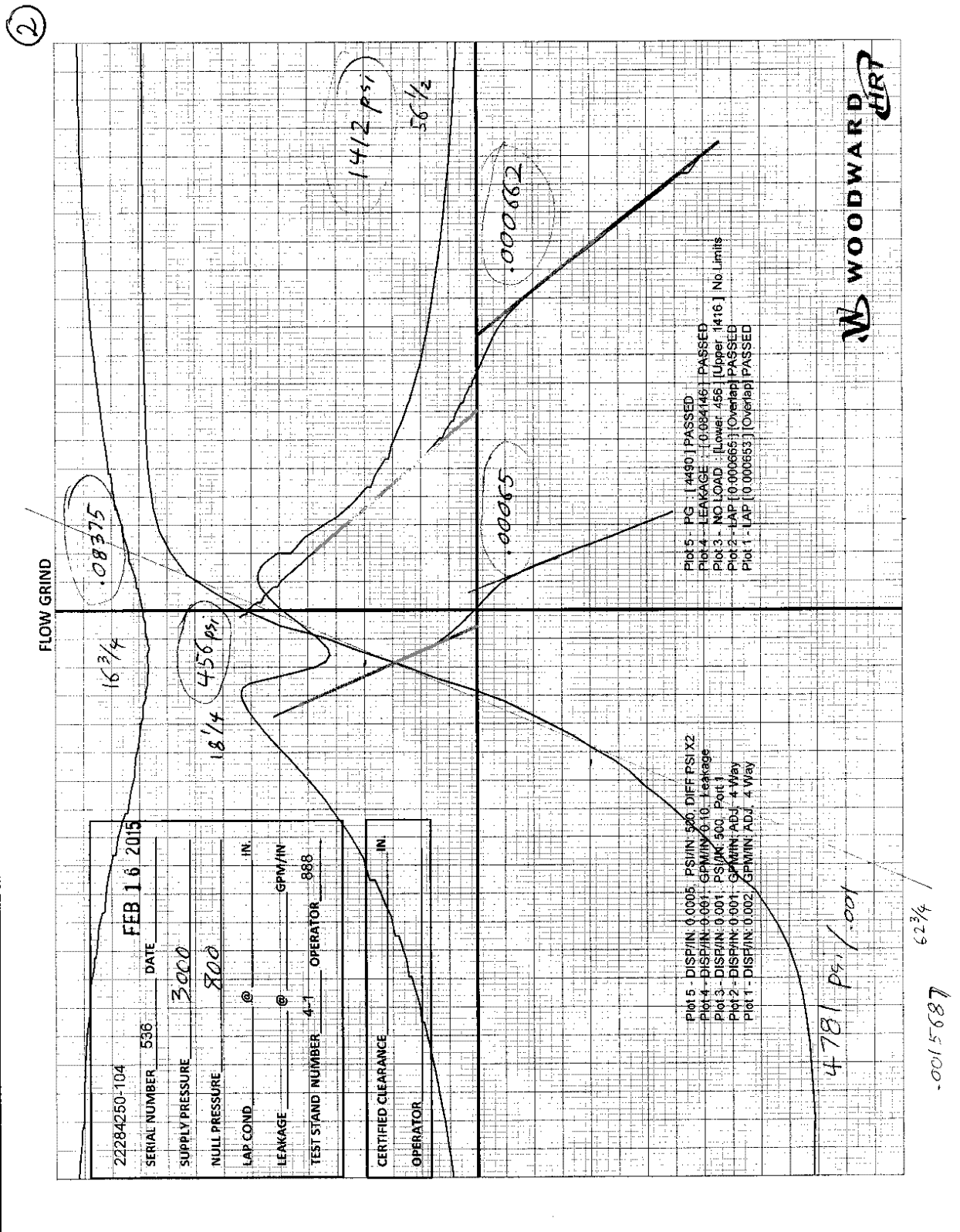
Appendix A-3

Servovalve 22284250-104

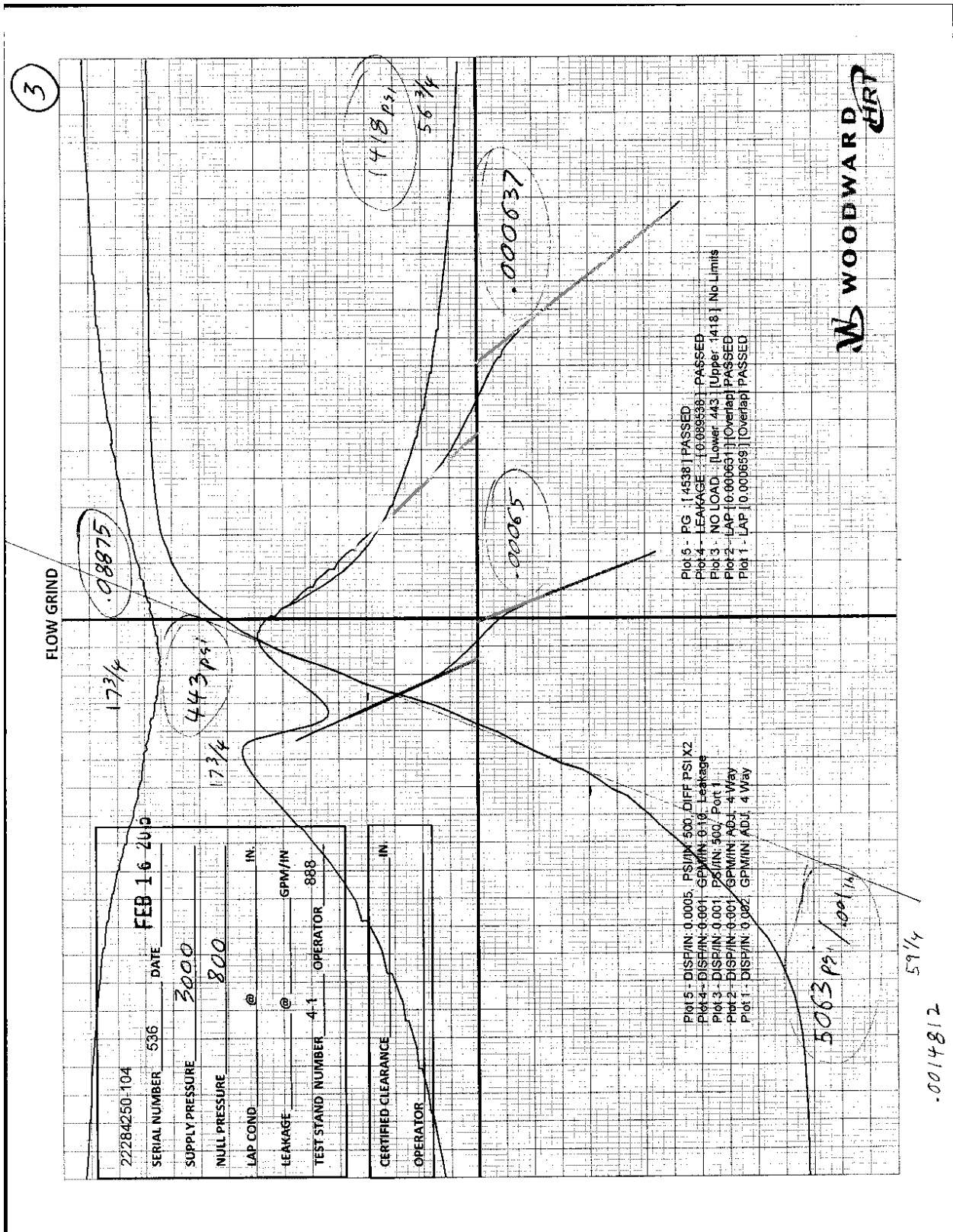
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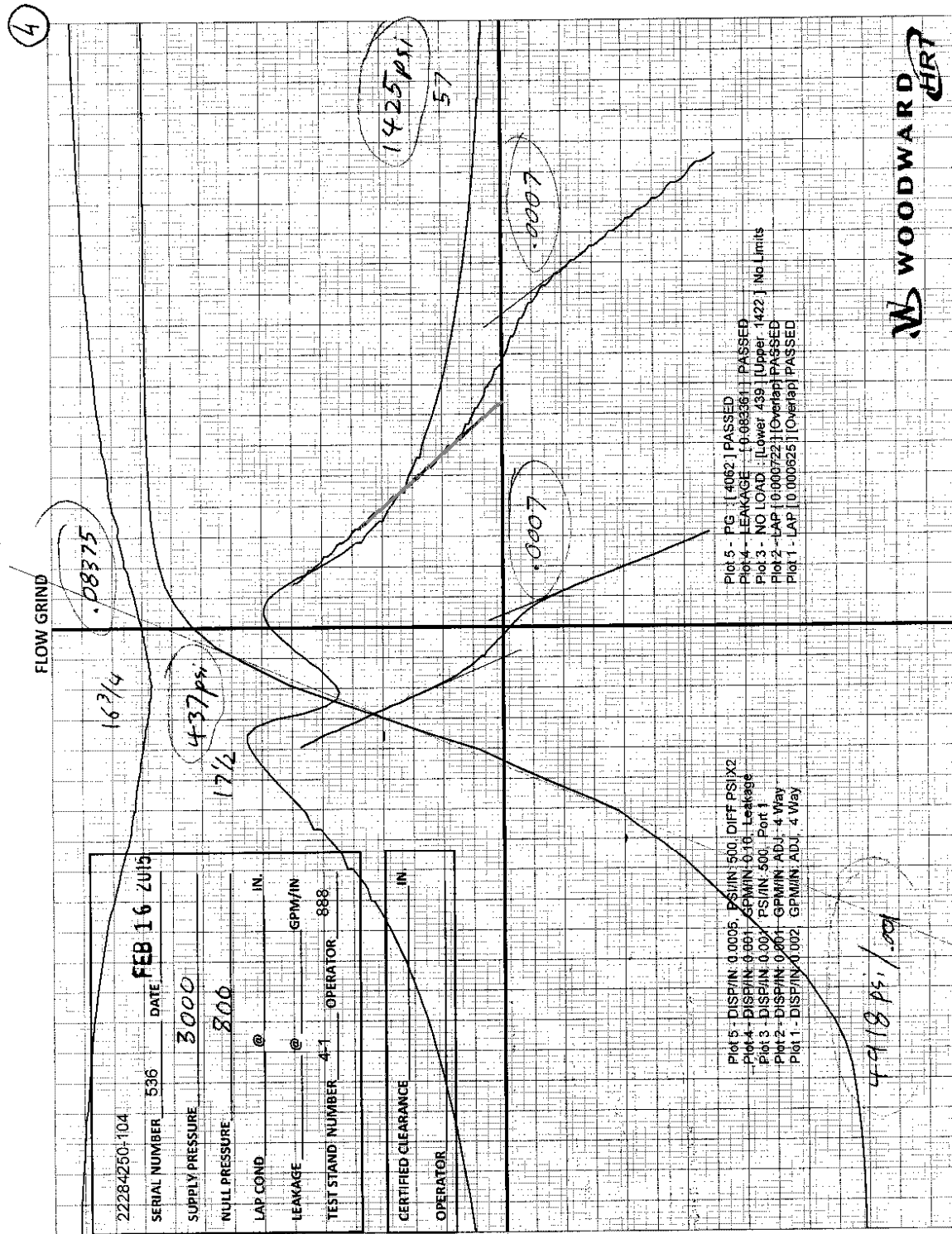


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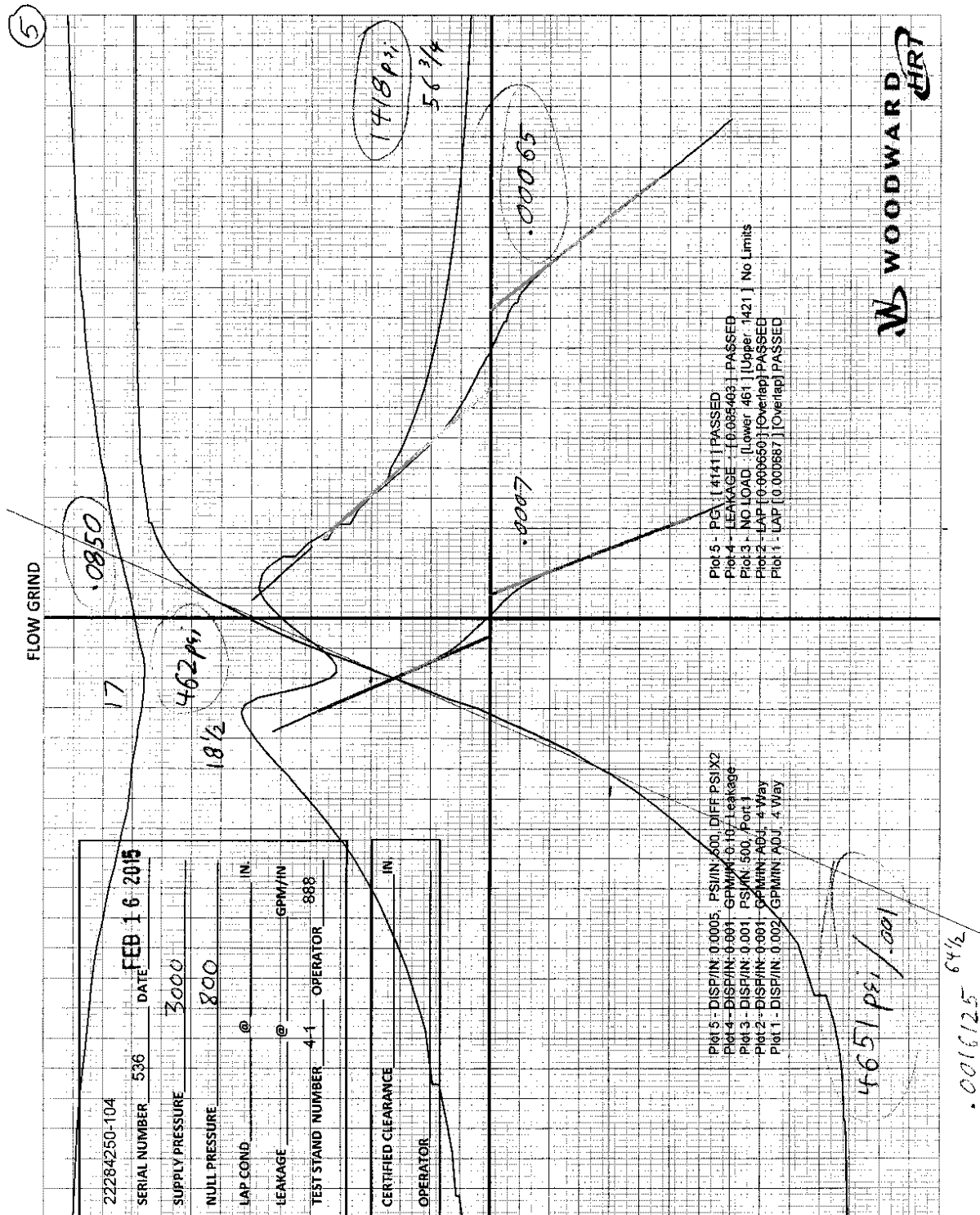


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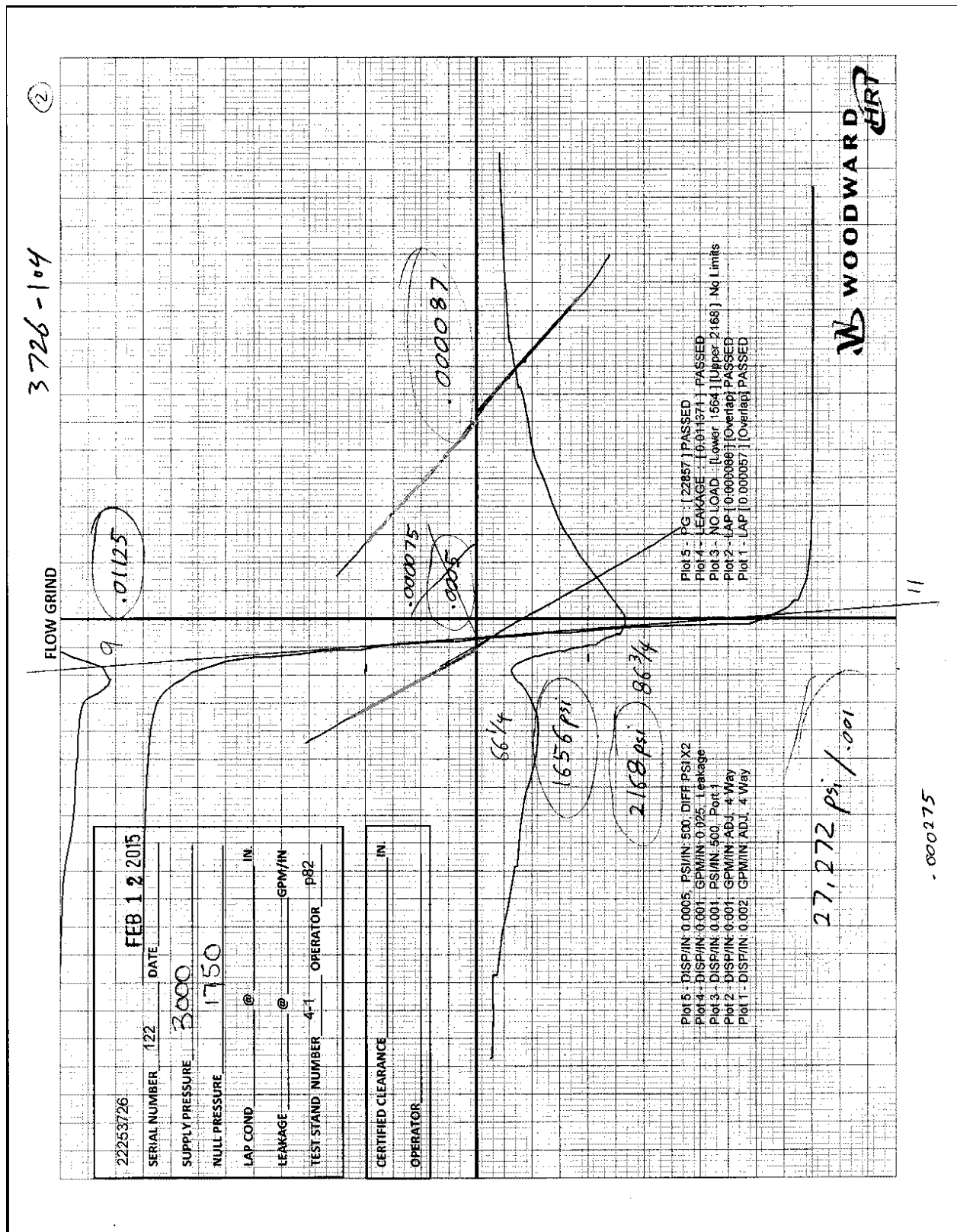


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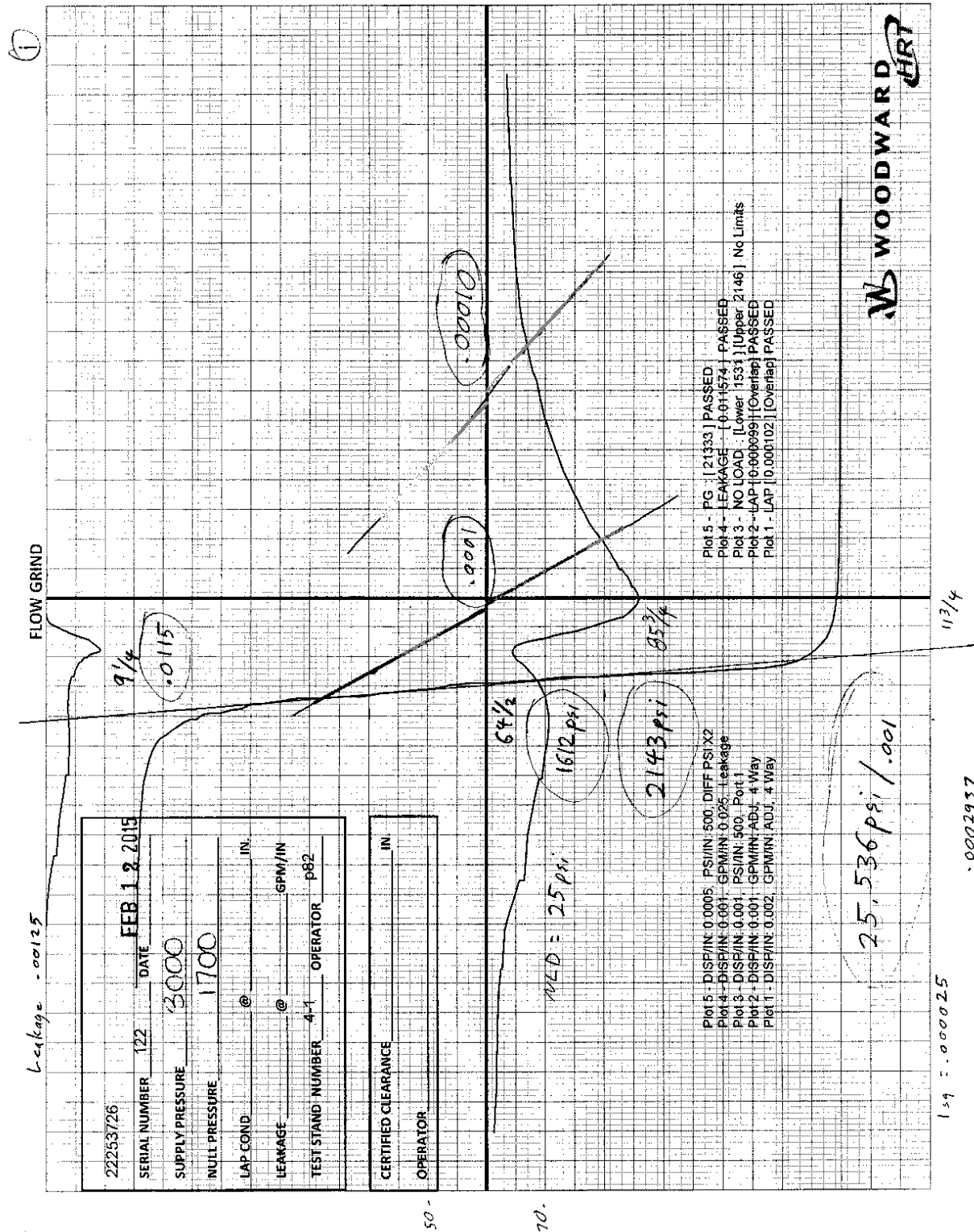
Appendix A-4

Servo valve 22253726 (25A)

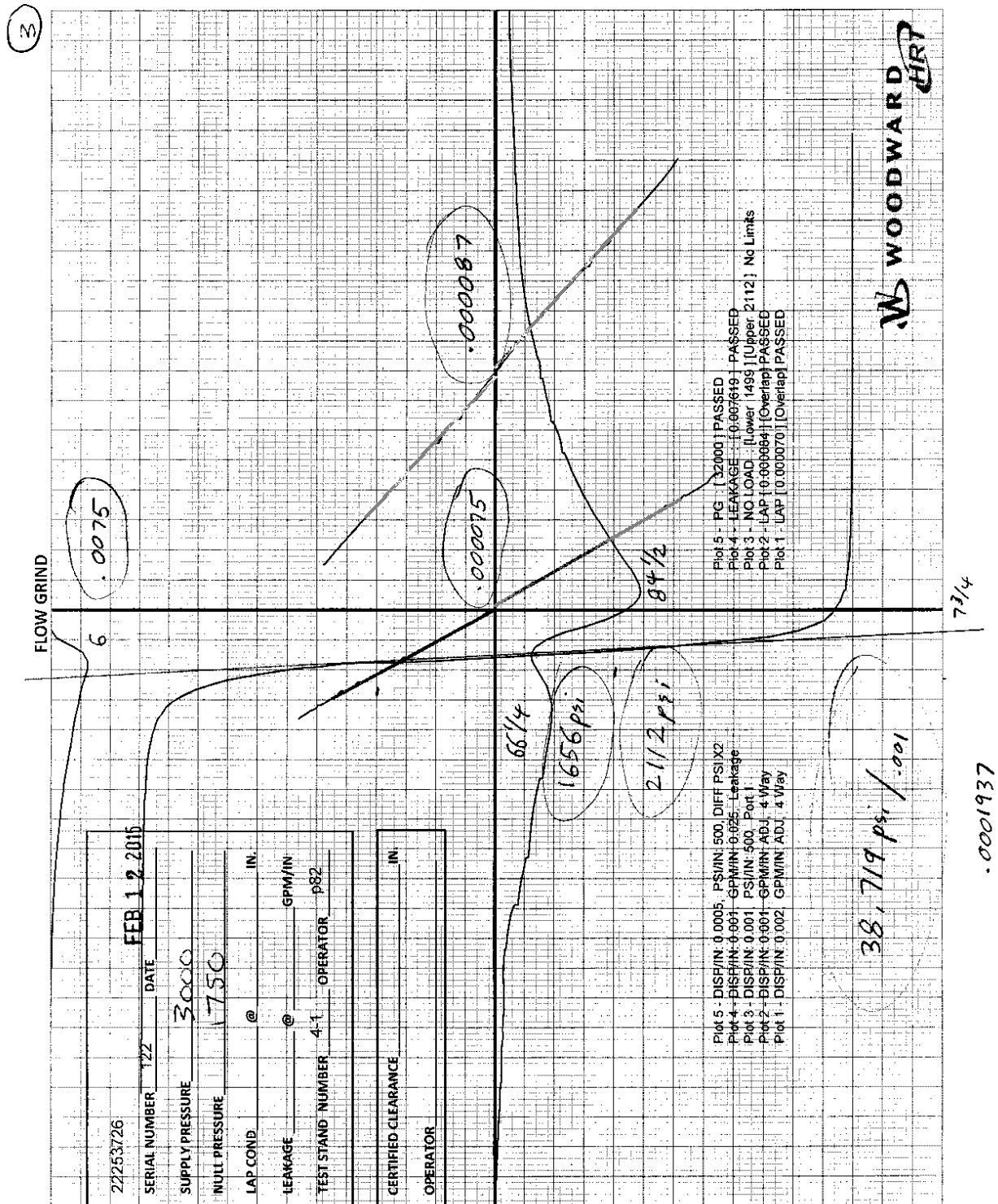
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